

Original Zeta, Full Support, and the Arithmetic Trace

Exact heat, divisor and matrix reconstruction

Split-Zero research programme

23 September 2026

This continuation keeps the original Riemann zeta function and every factor in its comparison with the theta heat family. It reconstructs the exceptional local fibres, the full signed divisor and its finite matrix receivers. In particular a calculated negative value of the uncompensated full divisor is accompanied by its calculated positive multiplier contribution. These different forms are related by the proved formulas; no RH proof or disproof is asserted. The full support lattice and all its prime stalks are derived first. A faithful labelled heat source then gives a bounded supported-zero action on its exact graph completion. Its two prime-boundary moments evaluate as explicit theta values and return the original zeta function. The final section constructs both sectorial continuations of the unfiltered endpoint and calculates the exact boundary direction lost by the theta filter.

Every divisor contribution remains in the receiving equation
Original zeta: the full multiplier and its inverse have specified domains

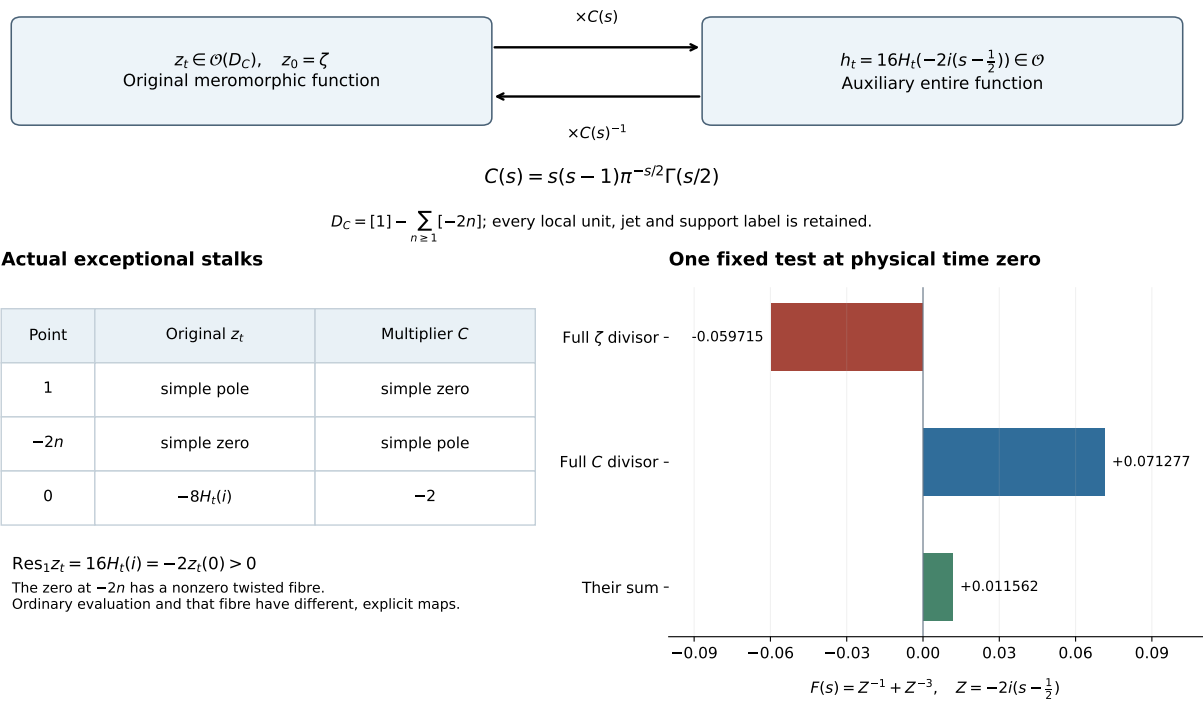


Figure 1: The actual multiplier and its line-module inverse (CF21), all exceptional real stalks (CF16, CF27–28), and a fixed original test at time zero (OZ8–13). Bar endpoints display midpoints of certified intervals; the outward bounds and all tails are in the proof. The heat source and constants are those of Rodgers–Tao, arXiv:1801.05914v5, equations `htdef`, `hoz` and `sas`. The two signed contributions are retained separately.

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1 Full-lattice primes, stalks, and retained factor divisors

Independent derivation, 2026-09-23. The actual bounded distributive support lattice is retained throughout. No location claim about zeta zeros follows from the spectrum alone.

1.1 Original sources read

The original [globalization note](#) supplies the two-label construction. Its relevant labels and source lines are thm:ideal-classification (207), cor:prime-ideals (239), cor:dim-two (262), thm:zero-prime-not-irr (298), thm:localization-away-from-zero (436), thm:frac-collapse (482), and cor:stalks (534).

The original [full-lattice geometry, version 11](#) supplies the lattice construction. Its locators are def:lattice-split (1229–1252), thm:lattice-normal-form (1255–1308), thm:lattice-localization (1457–1489), thm:lattice-ideal-classification (1619–1658), thm:lattice-prime-classification (1662–1701), thm:spectral-ordinal-sum (1710–1747), cor:lattice-split-dimension (1750–1769), branch construction (1877–1925), and thm:branch-selected-spectrum (1929–1973). The arithmetic retraction is at prop:classical-collapse (4653–4675), and the vector support construction at prop:split-anomaly-line (7573–7602).

The required algebra is fully proved below. These citations identify the original versions and exact source sections used; the results are rederived on the full lattice.

1.2 LP1. The actual semiring and universal maps

Let R be a nonzero commutative unital ring and L a nontrivial bounded distributive lattice, with bottom 0_L , top 1_L , join $\vee$, and meet $\wedge$. Set

$$S = G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\}, \quad (\text{LP1.1})$$

$$(r, \lambda) + (s, \mu) = (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu). \quad (\text{LP1.2})$$

Write $z_\lambda = (0, \lambda)$, $\tau = z_{0_L}$, $e = z_{1_L}$, and $\tilde{r} = (r, 1_L)$. Thus $\tilde{0} = e \neq \tau$. A nonzero amplitude requires top support. Closure holds because a nonzero sum has a top-support summand and a nonzero product has two top-support factors. Associativity and commutativity are componentwise, and distributivity is the ring and lattice distributive laws. The additive identity is τ , and the multiplicative unit is $\tilde{1}$.

Both projections are semiring maps:

$$p_L : S \rightarrow R, \quad (r, \lambda) \mapsto r, \quad \chi_L : S \rightarrow L, \quad (r, \lambda) \mapsto \lambda. \quad (\text{LP1.3})$$

Put $Z_L = \{z_\lambda : \lambda \in L\}$. Every semiring map from S to a ring kills Z_L , because $e + z_\lambda = e$ and additive cancellation in the target force the image of z_λ to be zero. Its restriction to the supported R is then a unital ring map. Conversely every such ring map extends by killing Z_L . This proves the universal property of p_L .

The units are exactly $\tilde{u}$, $u \in R^\times$: equality of a product with $\tilde{1}$ forces the amplitudes to multiply to 1; conversely the ring inverse at top support is a semiring inverse. In particular e is a nonunit.

1.3 LP2. All ideals, all primes, and their order

For a proper lattice ideal $\mathfrak{a}$ and ring ideal J , define

$$Z_{\mathfrak{a}} = \{z_\lambda : \lambda \in \mathfrak{a}\}, \quad I_J = Z_L \cup \{\tilde{r} : r \in J\}. \quad (\text{LP2.1})$$

A lattice ideal contains bottom, is lower, and is closed under finite joins. Every semiring ideal has exactly one of these forms. If it contains $\tilde{r}$, multiplication by $\tilde{-1}$ and addition put e in it, and $e z_\lambda = z_\lambda$ puts all of Z_L in it. Its top-support amplitudes form a ring ideal by addition and scalar absorption. If it contains no top-support element, it lies in Z_L and omits e . Its labels are join-closed, and $\mu \leq \lambda$ gives $z_\mu = z_\mu z_\lambda$, proving lower closure. Conversely the displayed sets are ideals by LP1.2. Their parameters are uniquely recovered, and the families are disjoint because only the second contains e .

A lattice prime is a proper lattice ideal for which membership of $\lambda \wedge \mu$ implies membership of one factor. The prime ideals of S are exactly

$$P_{\mathfrak{p}} = Z_{\mathfrak{p}} \quad (\mathfrak{p} \in \text{Spec}_{\text{lat}} L), \quad Q_{\mathfrak{q}} = I_{\mathfrak{q}} \quad (\mathfrak{q} \in \text{Spec } R). \quad (\text{LP2.2})$$

For a support ideal, two top-support factors have top-support product even when its amplitude vanishes. A zero-fiber factor multiplied by top support remains that factor. Two zero-fiber factors multiply by meet. These three cases prove exactly lattice primality. For an amplitude ideal, a product with a zero-fiber factor already has a factor in the ideal; top-support products give exactly ring primality. LP2.1 proves exhaustion.

The full order is

$$\begin{aligned} P_{\mathfrak{p}} \subseteq P_{\mathfrak{p}'} &\iff \mathfrak{p} \subseteq \mathfrak{p}', \\ Q_{\mathfrak{q}} \subseteq Q_{\mathfrak{q}'} &\iff \mathfrak{q} \subseteq \mathfrak{q}', \\ P_{\mathfrak{p}} \subsetneq Q_{\mathfrak{q}} &\text{ for all } \mathfrak{p}, \mathfrak{q}. \end{aligned} \quad (\text{LP2.3})$$

Read labels and amplitudes for the first two statements. The last is strict because the amplitude prime contains e .

1.4 LP3. Supported-zero principal prime and dimension

Multiplication and join closure give

$$(z_\lambda) = z_\lambda S = \{z_\mu : \mu \leq \lambda\}, \quad (e) = Z_L. \quad (\text{LP3.1})$$

Thus

$$(e) \text{ is prime} \iff R \text{ is an integral domain.} \quad (\text{LP3.2})$$

For a domain, divisibility by e is membership in this prime ideal, so $e \mid ab$ forces $e \mid a$ or $e \mid b$. But $e = e^2$ factors into two nonunits, so e is not irreducible. For $\lambda < 1_L$, (z_λ) is prime exactly when $\downarrow \lambda$ is a prime lattice ideal. In particular (τ) is prime exactly when bottom is meet-prime, which fails for two disjoint nonempty masks in a power-set lattice.

Write $h = \dim \text{Spec}_{\text{lat}} L$, defined as the supremum of finite strict chain lengths, possibly infinite. This spectrum is nonempty under the usual axiom of choice. Indeed Zorn gives a maximal proper lattice ideal. If $a \wedge b$ belongs to it but neither a nor b does, maximality supplies u, v in it with $u \vee a = v \vee b = 1_L$. Distributivity gives

$$1_L = (u \vee a) \wedge (v \vee b) \leq u \vee v \vee (a \wedge b),$$

contradicting properness. Hence that ideal is prime.

For a domain R ,

$$\text{ht}_S(e) = h + 1, \quad \text{ht}_S Q_{\mathfrak{q}} = h + 1 + \text{ht}_R \mathfrak{q}, \quad \dim S = h + 1 + \dim R. \quad (\text{LP3.3})$$

Each finite prime chain is a support chain followed by an amplitude chain, with one possible transition. Conversely every two chosen finite chains concatenate with one strict transition. Below $Q_{(0)}$ the amplitude part consists only of that point. Taking suprema proves every formula, including infinite values. For $R = \mathbb{Z}$, the dimension is $h + 2$. For $L = C_n = \{0 < 1 < \dots < n\}$, its n proper nonempty initial segments are exactly its prime ideals, and form a chain of length $n - 1$. Therefore

$$\dim G_{C_n}(\mathbb{Z}) = n + 1, \quad \text{ht}(e) = n. \quad (\text{LP3.4})$$

This gives arbitrarily large height for the principal supported-zero prime while preserving the arithmetic coefficient ring.

1.5 LP4. Arithmetic closed stratum

Prime membership gives

$$D_S(z_\lambda) = \{P_{\mathfrak{p}} : \lambda \notin \mathfrak{p}\}, \quad D_S(\tilde{r}) = \text{Spec}_{\text{lat}} L \cup D_R(r). \quad (\text{LP4.1})$$

These basic opens prove the ordinal-sum topology and

$$D_S(e) = \text{Spec}_{\text{lat}} L, \quad V_S(e) = \{Q_{\mathfrak{q}} : \mathfrak{q} \in \text{Spec } R\}. \quad (\text{LP4.2})$$

The reflection induces the closed embedding

$$i : \text{Spec } R \rightarrow \text{Spec } S, \quad \mathfrak{q} \mapsto Q_{\mathfrak{q}}. \quad (\text{LP4.3})$$

The quotient congruence is exactly equality of amplitudes: the Bourne relation $x + a = y + b$, $a, b \in Z_L$, implies equal amplitudes; conversely equal amplitudes give $x + e = y + e$. Hence this quotient is R .

For a domain there is a topological retraction

$$r(P_{\mathfrak{p}}) = (0), \quad r(Q_{\mathfrak{q}}) = \mathfrak{q}, \quad ri = \text{id}. \quad (\text{LP4.4})$$

The inverse image of $D_R(a)$ for $a \neq 0$ is $D_S(\tilde{a})$; that of $D_R(0)$ is empty. This proves continuity. The closure of (e) is precisely the arithmetic closed stratum.

This retraction cannot lift to a unital semiring-space retraction: its map at $Q_{(0)}$ would require a unital semiring map $K \rightarrow G_L(K)$, where $K = \text{Frac } R$. Composing any such map with χ_L and applying it to $1 + (-1) = 0$ gives $1_L \vee \lambda = 0_L$, impossible. The exact surviving bridges are the quotient embedding LP4.3 with its stalk maps below, and the proved topological retraction LP4.4.

1.6 LP5. All prime stalks with full lattice

Semiring fractions satisfy

$$(a, s) \sim (b, t) \iff \exists u \text{ in the denominators : } uat = ubs. \quad (\text{LP5.1})$$

The complement of $Q_{\mathfrak{q}}$ consists of $\tilde{s}$, $s \notin \mathfrak{q}$. There is an isomorphism

$$S_{Q_{\mathfrak{q}}} \longrightarrow G_L(R_{\mathfrak{q}}), \quad \frac{(a, \lambda)}{\tilde{s}} \mapsto (a/s, \lambda). \quad (\text{LP5.2})$$

It preserves operations, is onto, and equality in the target means equal labels and $u(at - bs) = 0$ for some $u \notin \mathfrak{q}$. These are precisely LP5.1, proving injectivity and well-definedness. If a nonzero numerator becomes zero in the localized ring, it retains its top label and becomes e .

For a domain this yields

$$\boxed{S_{(e)} \cong G_L(\text{Frac } R)}. \quad (\text{LP5.3})$$

Every lattice label survives. By contrast,

$$\boxed{S[e^{-1}] \cong L}. \quad (\text{LP5.4})$$

An invertible idempotent is the unit, so any map inverting e identifies x with $ex = z_{\chi_L(x)}$ and factors uniquely through χ_L , which itself inverts e . The same argument gives $S[z_{\lambda}^{-1}] \cong \downarrow \lambda$, using LP3.1 and the idempotent z_{λ} as the new unit.

At $P_{\mathfrak{p}}$, e belongs to the denominator set. First apply LP5.4, then invert $F = L \setminus \mathfrak{p}$:

$$\boxed{S_{P_{\mathfrak{p}}} \cong L_{\mathfrak{p}} = F^{-1}L}. \quad (\text{LP5.5})$$

Every fraction is represented by a label, with exact equivalence

$$\lambda \equiv_{\mathfrak{p}} \mu \iff \exists f \in F : \lambda \wedge f = \mu \wedge f. \quad (\text{LP5.6})$$

Every inverted idempotent becomes 1. The displayed relation is reflexive with witness 1_L , symmetric, transitive using the meet of witnesses, and compatible with join and meet by distributivity. It identifies every $f \in F$ with 1_L ; any map doing this respects the displayed relation. This proves its universal property.

A class equals 1 exactly when its label is outside $\mathfrak{p}$: use the label itself as witness in one direction; in the other, $\lambda \wedge f = f$ with $\lambda \in \mathfrak{p}, f \in F$ contradicts lower closure. The quotient remains a lattice and has only one unit, its top, because $x \wedge y = 1$ forces $x = y = 1$. Its unique maximal ideal is the image of $\mathfrak{p}$.

For finite L , put $f_{\mathfrak{p}} = \bigwedge F \in F$. Then

$$L_{\mathfrak{p}} \cong \downarrow f_{\mathfrak{p}}, \quad [\lambda] \mapsto \lambda \wedge f_{\mathfrak{p}}. \quad (\text{LP5.7})$$

Equality after meet with a witness $f \in F$ implies equality after meet with $f_{\mathfrak{p}} \leq f$, and the converse uses $f_{\mathfrak{p}}$. Every label below $f_{\mathfrak{p}}$ occurs. For $L = \{0 < a < 1\}$, the prime $\{0, a\}$ has stalk all of L , since its complement is $\{1\}$. Thus support stalks are not generally Boolean.

1.7 LP6. Stalk arrows and explicit branches

At the arithmetic closed stratum, the local reflection is

$$G_L(R_q) \rightarrow R_q, \quad (a, \lambda) \mapsto a. \quad (\text{LP6.1})$$

It is local because its inverse image of ring units is precisely the units identified in LP1. Its zero fiber is the complete L . The contravariant maps along every prime inclusion are

$$\begin{aligned} Q_q &\subseteq Q_{q'} : G_L(R_{q'}) \rightarrow G_L(R_q), & (a/s, \lambda) &\mapsto (a/s, \lambda), \\ P_p &\subseteq P_{p'} : L_{p'} \rightarrow L_p, & [\lambda]_{p'} &\mapsto [\lambda]_p, \\ P_p &\subsetneq Q_q : G_L(R_q) \rightarrow L_p, & (a, \lambda) &\mapsto [\lambda]_p. \end{aligned} \quad (\text{LP6.2})$$

The first two increase denominator sets. The last is support projection followed by LP5.6, and every arithmetic denominator maps to the unit. Thus each is the exact localization map; all triangles commute.

An explicit bounded lattice homomorphism $\epsilon : L \rightarrow \mathbf{B}$, and only such a selected branch in this construction, gives

$$\beta_\epsilon : G_L(R) \rightarrow G_{\mathbf{B}}(R), \quad z_\lambda \mapsto z_{\epsilon(\lambda)}, \quad \tilde{r} \mapsto \tilde{r}. \quad (\text{LP6.3})$$

Join and meet preservation prove the homomorphism. Taking preimages of the primes in LP2.2 shows that its spectral image is the selected $P_{\epsilon^{-1}(0)}$ together with all Q_q . No branch was selected in the preceding results.

1.8 LP7. Analytic local rings and full meromorphic germs

Let Y be a nonsingular connected complex curve, $y \in Y$, $\mathcal{O}_y$ its holomorphic local ring, $\mathfrak{m}_y$ its maximal ideal, and $\mathcal{M}_y = \text{Frac } \mathcal{O}_y$. The integral coefficient map induces

$$G_L(\mathbb{Z}) \rightarrow G_L(\mathcal{O}_y), \quad (n, \lambda) \mapsto (n \cdot 1, \lambda). \quad (\text{LP7.1})$$

It preserves every label and commutes with amplitude reflection.

For a chosen local parameter w at y , the first nonzero Taylor coefficient gives the exact full factorization

$$f = w^n u, \quad n \geq 0, \quad u \in \mathcal{O}_y^\times \quad (\text{LP7.2})$$

for every nonzero holomorphic germ. Quotients give the same expression for meromorphic germs with $n \in \mathbb{Z}$. The entire unit u is retained. A new parameter $w' = wv$, v a unit, gives $u' = v^{-n}u$; this proves n is intrinsic and proves the full change formula. Multiplication adds n and multiplies the full units. Hence

$$1 \rightarrow \mathcal{O}_y^\times \rightarrow \mathcal{M}_y^\times \xrightarrow{\text{ord}_y} \mathbb{Z} \rightarrow 0 \quad (\text{LP7.3})$$

is exact: its kernel is the retained unit group, and w^n attains each integer. The original germs are kept in the source; the order image does not replace them.

A nonzero ideal of $\mathcal{O}_y$ has a least order n ; an element of that order is w^n times a unit, so the ideal is (w^n) . Thus its only nonzero prime is $\mathfrak{m}_y = (w)$. The full lattice prime chain and its arithmetic stalks are

$$P_p \subsetneq Q_{(0)} = (e) \subsetneq Q_{\mathfrak{m}_y}, \quad (\text{LP7.4})$$

$$(G_L(\mathcal{O}_y))_{Q_{(0)}} = G_L(\mathcal{M}_y), \quad (G_L(\mathcal{O}_y))_{Q_{\mathfrak{m}_y}} = G_L(\mathcal{O}_y). \quad (\text{LP7.5})$$

Here equality denotes the explicit isomorphisms LP5.2. Evaluation

$$G_L(\mathcal{O}_y) \rightarrow G_L(\mathbb{C}), \quad (f, \lambda) \mapsto (f(y), \lambda) \quad (\text{LP7.6})$$

has amplitude-zero inverse image $Q_{\mathfrak{m}_y}$. A nonzero germ vanishing at y evaluates to e , with top support; before evaluation LP7.2 retains its exact order and full unit. A pole is a meromorphic germ with negative order. The element e is not a parameter: $e^2 = e$. Equations LP7.3–LP7.6 give its precise relation to meromorphic localization and analytic orders.

1.9 LP8. Full factor source and tagged divisor module

Let $\mathcal{A}$ be a nonempty finite set of original factor labels and let F_α , $\alpha \in \mathcal{A}$, be specified nonzero meromorphic functions on Y . Every original constant, power, and factor remains in this tuple, including factors with zero divisor. Put

$$D = \bigcup_{\alpha \in \mathcal{A}} \{y : \text{ord}_y F_\alpha \neq 0\}. \quad (\text{LP8.1})$$

The set is locally finite by isolation of zeros and poles and finiteness of $\mathcal{A}$. Points at which the product has net order zero are retained. At every $y \in Y$, including outside D , the entire original germ tuple in $(\mathcal{M}_y^\times)^\mathcal{A}$ remains available.

With $S_\mathbb{Z} = G_L(\mathbb{Z})$, define

$$\mathcal{T}_L(U) = \prod_{y \in D \cap U} S_\mathbb{Z}^\mathcal{A}, \quad \mathcal{C}_L(U) = \prod_{y \in D \cap U} S_\mathbb{Z}. \quad (\text{LP8.2})$$

Restrictions delete coordinates outside a smaller open. Coordinatewise operations make these sheaves of semimodules: compatible sections glue uniquely at each coordinate. Their stalks at $y \in D$ are $S_\mathbb{Z}^\mathcal{A}$, $S_\mathbb{Z}$, since a sufficiently small neighborhood contains no other divisor point; away from D the stalk is zero. Products allow globally infinite divisors without dropping their locally finite contributions.

Define integral amplitude sheaves

$$\mathcal{D}_{\text{tag}}(U) = \prod_{y \in D \cap U} \mathbb{Z}^\mathcal{A}, \quad \mathcal{D}(U) = \prod_{y \in D \cap U} \mathbb{Z}. \quad (\text{LP8.3})$$

Amplitude and support give an injection

$$\mathcal{T}_L(U) \hookrightarrow \mathcal{D}_{\text{tag}}(U) \times \prod_{y \in D \cap U} L^\mathcal{A}. \quad (\text{LP8.4})$$

Its image is exactly the pairs satisfying $n_{\alpha,y} \neq 0 \Rightarrow \lambda_{\alpha,y} = 1_L$. This is LP1.1 coordinatewise; in particular the amplitude-zero fiber is the entire product of lattice labels.

The supplied factor tuple maps to

$$(\mathbf{d}_F)_{\alpha,y} = (\text{ord}_y F_\alpha, 1_L). \quad (\text{LP8.5})$$

Every supplied factor has an entry at every point of D . An order-zero factor has value e , not absence. The complete functions and unit germs are retained separately in the source tuple. The exact local map is

$$\begin{array}{ccc} (\mathcal{M}_y^\times)^\mathcal{A} & \xrightarrow{\Pi_\alpha} & \mathcal{M}_y^\times \\ \downarrow (\text{ord}_y)_\alpha & & \downarrow \text{ord}_y \\ \mathbb{Z}^\mathcal{A} & \xrightarrow{\Sigma_\alpha} & \mathbb{Z}. \end{array} \quad (\text{LP8.6})$$

Both vertical maps are onto, with kernels $(\mathcal{O}_y^\times)^\mathcal{A}$ and $\mathcal{O}_y^\times$, by LP7.3. The square commutes because orders add. It specifies the information carried by the divisor image and the full unit information remaining in the source. The supplied top-support lift of an order tuple is not claimed to preserve additive zero: a supplied zero order intentionally maps to e .

1.10 LP9. Exact cancellation and supported fibers

Finite tag sum is the semimodule map

$$\Delta : \mathcal{T}_L \rightarrow \mathcal{C}_L, \quad ((n_\alpha, \lambda_\alpha))_\alpha \mapsto \left(\sum_\alpha n_\alpha, \bigvee_\alpha \lambda_\alpha \right). \quad (\text{LP9.1})$$

Additivity follows from associativity and commutativity. Scalar compatibility follows from integer distributivity and $\bigvee_{\alpha}(\mu \wedge \lambda_{\alpha}) = \mu \wedge \bigvee_{\alpha} \lambda_{\alpha}$. Thus Δ is $S_{\mathbb{Z}}$ -linear. It is onto by putting a target in one tag and τ in all others. The amplitude square with the ordinary sum $\Sigma : \mathcal{D}_{\text{tag}} \rightarrow \mathcal{D}$ commutes coordinatewise. For the supplied tuple,

$$\Delta(\mathbf{d}_F)_y = \left(\text{ord}_y \prod_{\alpha} F_{\alpha}, 1_L \right). \quad (\text{LP9.2})$$

Cancellation gives $e \neq \tau$, while every original factor order remains in $\mathbf{d}_F$ and every full factor remains in LP8.6.

The exact fibers at a divisor point are

$$\begin{aligned} \Delta^{-1}(e) &= \{(n_{\alpha}, \lambda_{\alpha}) : \sum_{\alpha} n_{\alpha} = 0, \bigvee_{\alpha} \lambda_{\alpha} = 1_L\}, \\ \Delta^{-1}(\tau) &= \{(\tau)_{\alpha}\}. \end{aligned} \quad (\text{LP9.3})$$

The first is equality of the two output coordinates. For the second, a finite join equals bottom only when all entries are bottom, and each bottom label forces amplitude zero. The full congruence is equality of the sums and equality of the joins; it is not determined merely by the additive-zero preimage.

For $k = |\mathcal{A}|$, the canceled integer subgroup has the complete split exact sequence

$$0 \rightarrow \mathbb{Z}^{k-1} \xrightarrow{(a_1, \dots, a_{k-1}) \mapsto (a_1, \dots, a_{k-1}, -\sum_{j < k} a_j)} \mathbb{Z}^k \xrightarrow{\Sigma} \mathbb{Z} \rightarrow 0. \quad (\text{LP9.4})$$

The first map is injective with exactly the zero-sum image, and the last is split by its last coordinate. For $k = 1$ the first group is zero. This identifies the canceled integer data and relates them to the supported fiber without deleting the tags.

1.11 LP10. Tagged source at the actual coefficient primes

At $y \in D$, the fiber is finite free. For every coefficient prime P ,

$$(S_{\mathbb{Z}}^{\mathcal{A}})_P \cong (S_{\mathbb{Z}, P})^{\mathcal{A}}. \quad (\text{LP10.1})$$

The map is coordinatewise. Multiply the finitely many denominators to get a common denominator for surjectivity, and multiply the finitely many equality witnesses from LP5.1 for injectivity. No commutation with an infinite product of global sections is asserted. Thus the three kinds of fibers are

$$\begin{aligned} P = (e) : & \quad G_L(\mathbb{Q})^{\mathcal{A}}, \\ P = Q_{(p)} : & \quad G_L(\mathbb{Z}_{(p)})^{\mathcal{A}}, \\ P = P_{\mathfrak{p}} : & \quad L_{\mathfrak{p}}^{\mathcal{A}}. \end{aligned} \quad (\text{LP10.2})$$

The tag-sum map commutes with localization by linearity. The maps from $G_L(\mathbb{Z})$ to $G_L(\mathbb{Q})$ and $G_L(\mathbb{Z}_{(p)})$ are injective, because both preserve labels and their ring maps reflect equality of integers. Consequently every tagged signed integer order and every canceled integer vector survives at the supported-zero and arithmetic prime stalks, and e remains distinct from τ .

At a support prime the exact map is $(n, \lambda) \mapsto [\lambda]_{\mathfrak{p}}$. Each supplied top-support order becomes top independently in each tag. This is the exact support image; the original factor source and arithmetic stalks retain the integer and germ data through the proved maps.

1.12 LP11. Scope for original zeta

The base contributes the full meromorphic stalk at the supported-zero prime, the arithmetic amplitude quotient, all support stalks, the local valuation sequence with its retained unit kernel,

and a tagged source whose cancellation map sends zero sums to e . These are exact maps available to the original zeta and completion-factor calculations.

They do not alone force zero locations. For example, in the original complex coordinate s , translation gives

$$\mathcal{O}_{\mathbb{C}, y+a} \rightarrow \mathcal{O}_{\mathbb{C}, y}, \quad F(s) \mapsto F(s+a). \quad (\text{LP11.1})$$

Its inverse is translation by $-a$. It preserves maximal ideals, extends to meromorphic fields, preserves each order and its full transported unit, and induces G_L -isomorphisms fixing all labels. Indeed $s+a-(y+a)=s-y$, so the full local factorization transports exactly. The tagged divisor point $y+a$ maps to y with every coefficient and factor label retained, and the cancellation square commutes. This is a comparison of local structures, not a coordinate substitution in an original zeta calculation. The actual original functions, completion factors, symmetries and differential terms supply the further analytic data.

The original-zeta trace and heat calculations receive this base through the exact stalk and factor maps above. Their analytic functions and differential terms remain additional specified data.

2 The original zeta function with its complete meromorphic factor retained

CF1. Original source, coordinates, and multiplier. The original author source is Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, version 5, equations `hoz`, `sas`, `phidef`, and `htdef`. The defining passage was read directly in its retained author TeX, lines 75–106; no claim of a complete reading is made here. It gives exactly

$$\begin{aligned} \Phi(u) &= \sum_{n=1}^{\infty} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \\ H_t(Z) &= \int_0^{\infty} e^{tu^2} \Phi(u) \cos(Zu) du, \\ H_0(Z) &= \frac{1}{8} \xi \left(\frac{1}{2} + \frac{iZ}{2} \right), \quad \xi(s) = \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2) \zeta(s). \end{aligned} \quad (\text{CF1})$$

Here ζ is the original Riemann zeta function, initially $\sum_{n \geq 1} n^{-s}$ on $\Re s > 1$, with its meromorphic continuation. Keep the precise receiving map

$$\begin{aligned} Z &= -2i(s - \tfrac{1}{2}), \quad h_t(s) = 16H_t(-2i(s - \tfrac{1}{2})), \\ C(s) &= s(s-1)\pi^{-s/2}\Gamma(s/2), \quad h_0(s) = 2\xi(s) = \underbrace{s(s-1)\pi^{-s/2}\Gamma(s/2)}_{C(s)} \zeta(s). \end{aligned} \quad (\text{CF2})$$

Every appearance of C below refers to this full meromorphic function. None of its four displayed factors is discarded.

The Gamma input is the original-author treatment of Ricardo Pérez-Marco, *On the definition of Euler Gamma function*, version 2, Theorem `thm:main-Euler-gamma`, Corollary `cor:W-product`, and Corollary `cor:Gamma(1/2)`. Its original TeX `definition-gamma-Euler6.tex` and intact source archive are retained. The direct reading covers lines 202–340, 1170–1205, and the Euler-integral theorem at 1332–1389. We also use the primary TeX equations in R. A. Askey and R. Roy’s [DLMF Chapter 5: 5.5.1, 5.7.3, 5.7.4, and 5.8.2](#). All four equation TeX files are retained and were read completely. Their required consequences are derived below.

CF2. Analyticity, positivity, and the exact heat clock. Every summand of $\Phi(u)$ is strictly positive for $u \geq 0$: its sign is the sign of $2\pi n^2 e^{4u} - 3 > 0$. There is a finite constant

$K_\Phi > 0$ such that

$$0 < \Phi(u) \leq K_\Phi e^{9u} e^{-\pi e^{4u}/2}, \quad u \geq 0. \quad (\text{CF3})$$

Indeed, each complete summand is at most $2\pi^2 n^4 e^{9u} e^{-\pi n^2 e^{4u}}$. For $E = e^{4u} \geq 1$, $n^2 E \geq E/2 + n^2/2$, so

$$\sum_{n \geq 1} n^4 e^{-\pi n^2 E} \leq e^{-\pi E/2} \sum_{n \geq 1} n^4 e^{-\pi n^2/2}.$$

This proves CF3 with $K_\Phi = 2\pi^2 \sum_{n \geq 1} n^4 e^{-\pi n^2/2}$. It does not change the kernel in CF1.

On a compact set of complex $t, \bar{Z}$, every mixed derivative of the integrand is bounded by a constant times

$$u^N \exp(Tu^2 + Bu + 9u - \pi e^{4u}/2)$$

for some finite N, T, B . This is integrable: e^{4u} eventually dominates every polynomial in u , and the integrand is bounded on finite intervals. Differentiation under the integral and locally uniform convergence therefore prove joint entire dependence on t, Z , and

$$\partial_t H_t = -\partial_{\bar{Z}}^2 H_t.$$

Since $(-2i)^2 = -4$, applying the actual map CF2 gives

$$\partial_t h_t(s) = \frac{1}{4} \partial_s^2 h_t(s). \quad (\text{CF4})$$

The factor $1/4$ is forced by the original arithmetic coordinate and is not a choice of heat clock.

For real t, y , the complete integral is

$$H_t(iy) = \int_0^\infty e^{tu^2} \Phi(u) \cosh(yu) du > 0. \quad (\text{CF5})$$

Evenness of the cosine gives $h_t(1-s) = h_t(s)$, including complex s, t . In particular, for every real t, s , $h_t(s) > 0$. None of the real exceptional points examined below is a zero of h_t .

CF3. Gamma poles and the divisor of the complete factor. The original-author Gamma product, also DLMF 5.8.2, is

$$\frac{1}{\Gamma(w)} = w e^{\gamma w} \prod_{j=1}^\infty (1 + w/j) e^{-w/j}. \quad (\text{CF6})$$

On compact sets avoiding its displayed zeros, the logarithms of the tail factors converge locally uniformly because $\log(1 + w/j) - w/j = O(j^{-2})$. Thus the reciprocal is entire, with simple zeros precisely at $0, -1, -2, \dots$, and no other zeros. Consequently Γ has exactly the corresponding simple poles and has no zeros. Using $\Gamma(w+1) = w\Gamma(w)$ and $\Gamma(1) = 1$, its residues are

$$\text{Res}_{w=-n} \Gamma(w) = \frac{(-1)^n}{n!}, \quad n \geq 0. \quad (\text{CF7})$$

For a direct derivation, write $\Gamma(w) = \Gamma(w+n+1)/[w(w+1)\cdots(w+n)]$ near $w = -n$. The numerator tends to one, and the denominator divided by $w+n$ tends to $(-1)^n n!$.

Apply these facts to the original product $C(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)$. At $s = 1$, its simple-zero coefficient is

$$\lim_{s \rightarrow 1} \frac{C(s)}{s-1} = 1 \cdot \pi^{-1/2} \Gamma(1/2) = 1. \quad (\text{CF8})$$

Here $\Gamma(1/2) = \sqrt{\pi}$ is the cited Gamma corollary. At $a_n = -2n$, $n \geq 1$, the chain factor two in the Gamma residue gives

$$\kappa_n := \text{Res}_{s=a_n} [s(s-1)\pi^{-s/2}\Gamma(s/2)] = \frac{4n(2n+1)(-1)^n \pi^n}{n!}. \quad (\text{CF9})$$

Each factor in this calculation is retained: $a_n(a_n - 1) = 2n(2n + 1)$, $\pi^{-a_n/2} = \pi^n$, and the Gamma residue is $2(-1)^n/n!$. At zero, the factor s and the Gamma pole have opposite orders; the next paragraph calculates their full product. It has value -2 , so there is no zero or pole there. There are no other zeros or poles of C . Thus, as a locally finite divisor on the complex plane,

$$\operatorname{div}\left(s(s-1)\pi^{-s/2}\Gamma(s/2)\right) = [1] - \sum_{n \geq 1} [-2n] =: D_C. \quad (\text{CF10})$$

This is a divisor on $\mathbb{C}$; no meromorphic extension of C across a point at infinity is asserted.

CF4. The unreduced zero-point Laurent product. CF6 and the Gamma recurrence give, for $|w| < 1$,

$$\log \Gamma(1+w) = -\gamma w + \sum_{k \geq 2} \frac{(-1)^k \zeta(k)}{k} w^k. \quad (\text{CF11})$$

To prove it, multiply CF6 by $1/w$, use $\Gamma(1+w) = w\Gamma(w)$, and expand $\log(1+w/j)$. The interchange is justified on $|w| \leq r < 1$ by the convergent majorant $\sum_{k \geq 2} r^k \zeta(k)/k$. This agrees with the cited DLMF expansions; only the smaller disk $|w| < 1$ is needed here. Exponentiation and division by $w = s/2$ give

$$\Gamma(s/2) = \frac{2}{s} - \gamma + \frac{\gamma^2 + \zeta(2)}{4} s + O(s^2).$$

Write $L_\pi = \log \pi$. Keeping the separate original s and Gamma pole factors, the exact receiving expansion is

$$\begin{aligned} C(s) &= s(s-1) \left(1 - \frac{L_\pi s}{2} + \frac{L_\pi^2 s^2}{8} + O(s^3)\right) \left(\frac{2}{s} - \gamma + \frac{\gamma^2 + \zeta(2)}{4} s + O(s^2)\right) \\ &= -2 + (2 + \gamma + L_\pi)s \\ &\quad - \left(\gamma + L_\pi + \frac{(\gamma + L_\pi)^2 + \zeta(2)}{4}\right) s^2 + O(s^3). \end{aligned} \quad (\text{CF12})$$

The first line is the Laurent product of the actual four factors, not a substitution of a constant for their product. It proves $C(0) = -2$ and retains both finite derivatives.

Let $c = C'/C$, with C still the complete function in CF2. It is holomorphic near zero, and CF12 yields

$$c(0) = -1 - \frac{\gamma + \log \pi}{2}, \quad c'(0) = -1 + \frac{\zeta(2)}{4}. \quad (\text{CF13})$$

Indeed $c(0) = C'(0)/C(0)$ and $c'(0) = C''(0)/C(0) - [C'(0)/C(0)]^2$. Substituting the three coefficients in CF12 proves both identities. Away from the individual singular factors,

$$c(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{\log \pi}{2} + \frac{1}{2} \psi(s/2). \quad (\text{CF14})$$

The $1/s$ term and the pole part of $\psi(s/2)/2$ cancel at zero. CF12–13 give their actual continued value and derivative; the cancellation does not authorize removal of their stored source factors.

CF5. The transported original-zeta family and every real-time stalk. Define the meromorphic family by the exact comparison

$$z_t(s) = [s(s-1)\pi^{-s/2}\Gamma(s/2)]^{-1} 16H_t(-2i(s-\tfrac{1}{2})), \quad h_t = Cz_t. \quad (\text{CF15})$$

CF2 proves $z_0 = \zeta$ as meromorphic functions. Thus this is the transport of the original zeta function under the specified completed heat evolution, not an unrelated choice of initial value.

At every real t , CF5, CF8, CF9, and CF12 prove

$$\begin{aligned} \operatorname{Res}_{s=1} z_t(s) &= 16H_t(i) > 0, \\ z_t(0) &= -8H_t(i) < 0, \\ z'_t(-2n) &= \frac{16H_t(i(4n+1))}{\kappa_n} \neq 0, \quad n \geq 1, \\ \operatorname{Res}_{s=1} z_t &= -2z_t(0). \end{aligned} \tag{CF16}$$

For the first equality, $C/(s-1) \rightarrow 1$ and $h_t(1) = 16H_t(-i) = 16H_t(i)$. For the second, $C(0) = -2$ and $h_t(0) = 16H_t(i)$. For the third, $(s+2n)C(s) \rightarrow \kappa_n$, so

$$z_t(s) = \frac{h_t(-2n)}{\kappa_n}(s+2n) + O((s+2n)^2). \tag{CF17}$$

The numerator is strictly positive by CF5, and the argument in CF2 is exactly $i(4n+1)$. Consequently the pole at one and all zeros at negative even integers are genuine and simple for every real time. Zero is a regular nonzero point. The sign of $z'_t(-2n)$ is $(-1)^n$. These assertions evaluate the actual stalks, rather than leaving their multiplicities subject to an unknown cancellation with h_t .

For completeness the original values at time zero can be recovered directly. On $\Re s > 1$, integration against the counting function gives

$$\zeta(s) = s \int_1^\infty [x] x^{-s-1} dx = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx.$$

The last integral is holomorphic for $\Re s > 0$, including a neighborhood of one, because $0 \leq \{x\} < 1$ and derivatives introduce integrable powers of $\log x$. Thus $\operatorname{Res}_1 \zeta = 1$. CF16 then gives $16H_0(i) = 1$ and $\zeta(0) = -1/2$. The same argument fixes these constants without changing any factor in CF2.

CF6. The exact drift and potential, with their domains. On the open set

$$U_C = \mathbb{C} \setminus (\{1\} \cup \{-2, -4, \dots\})$$

the full $C = s(s-1)\pi^{-s/2}\Gamma(s/2)$ is a holomorphic nonvanishing function, with its removable value at zero included. Differentiate $h_t = Cz_t$; the factor is independent of t . CF4 yields

$$\begin{aligned} \partial_t z_t &= L_C z_t, \\ L_C f &:= [s(s-1)\pi^{-s/2}\Gamma(s/2)]^{-1} \frac{1}{4} \partial_s^2 [s(s-1)\pi^{-s/2}\Gamma(s/2) f] \\ &= \frac{1}{4} f'' + \frac{C'}{2C} f' + \frac{C''}{4C} f = \frac{1}{4} f'' + \frac{c}{2} f' + \frac{c' + c^2}{4} f. \end{aligned} \tag{CF18}$$

The product rule proves the identity with every coefficient retained. It is initially an identity of holomorphic functions on U_C . Each side extends meromorphically across the omitted points, so it is also an identity of meromorphic germs everywhere. The next paragraph proves its precise holomorphic line-module domain there.

At zero, no coefficient is singular. Its exact received equation is

$$\partial_t z_t|_{s=0} = \frac{1}{4} z_t''(0) + \frac{1}{2} \left(-1 - \frac{\gamma + \log \pi}{2} \right) z_t'(0) + \frac{1}{4} \left[\left(-1 - \frac{\gamma + \log \pi}{2} \right)^2 - 1 + \frac{\zeta(2)}{4} \right] z_t(0). \tag{CF19}$$

This is CF18 with the actual coefficients CF13, not the heat equation with its first- and zeroth-order terms removed.

CF7. Exact stalk domains and the full local units. Let $\mathcal{O}$ be the sheaf of holomorphic functions on $\mathbb{C}$ and $\mathcal{M}$ its sheaf of meromorphic functions. For the actual divisor CF10 define

$$\mathcal{O}(D_C)(V) = \{f \in \mathcal{M}(V) : \text{ord}_a f + \text{ord}_a C \geq 0 \text{ for all } a \in V\}. \quad (\text{CF20})$$

The zero function is included. There is an exact isomorphism of holomorphic line modules

$$\begin{aligned} \mu_C : \mathcal{O}(D_C) &\longrightarrow \mathcal{O}, & f &\longmapsto s(s-1)\pi^{-s/2}\Gamma(s/2)f, \\ \mu_C^{-1}(h) &= [s(s-1)\pi^{-s/2}\Gamma(s/2)]^{-1}h. \end{aligned} \quad (\text{CF21})$$

To prove this at every stalk, the condition in CF20 says exactly that Cf is holomorphic. Conversely h/C satisfies that condition whenever h is holomorphic. The two maps compose to the identity as meromorphic germs, including each exceptional stalk, and commute with restriction. In particular the actual z_t is a global section of this line module.

Here are its local domains with the complete units retained. For a point a , set $x = s - a$ and $k_a = \text{ord}_a C$. Define the actual holomorphic nonvanishing germ

$$u_a(x) = x^{-k_a}(a+x)(a+x-1)\pi^{-(a+x)/2}\Gamma((a+x)/2). \quad (\text{CF22})$$

This is the exact local factor identity $C(a+x) = x^{k_a}u_a(x)$. It records the full original multiplier, including every Taylor coefficient of u_a . The values are

a	k_a	$u_a(0)$	$\mathcal{O}(D_C)_a$
1	1	1	$x^{-1}\mathcal{O}_a$
$-2n, n \geq 1$	-1	κ_n	$x\mathcal{O}_a$
0	0	-2	$\mathcal{O}_0$.

(CF23)

The line on the right may be written more precisely with its actual generator as $x^{-k_a}u_a(x)^{-1}\mathcal{O}_a$; the equality with the stated fractional ideals uses that u_a is invertible and does not replace that generator. The full section identity is

$$z_t(a+x) = x^{-k_a}u_a(x)^{-1}h_t(a+x). \quad (\text{CF24})$$

At $a = 1$ it has at most a simple pole, at $a = -2n$ it vanishes at least once, and at zero it is holomorphic. CF16 proves the stronger exact orders for every real t .

The operator in CF18 is a well-defined differential operator

$$L_C : \mathcal{O}(D_C) \longrightarrow \mathcal{O}(D_C), \quad \mu_C L_C = \frac{1}{4}\partial_s^2 \mu_C, \quad C = s(s-1)\pi^{-s/2}\Gamma(s/2). \quad (\text{CF25})$$

Indeed $f = C^{-1}g$, $g \in \mathcal{O}$, gives $L_C f = C^{-1}g''/4$. This proves invariance of the domain, including at every exceptional point. It also proves that each apparent singular coefficient in CF18 is accounted for by the precise line module rather than by an unrecorded cancellation rule.

For an explicit local check put $a_u = u'_a/u_a$. Then the full multiplier CF22 gives

$$\begin{aligned} c &= \frac{k_a}{x} + a_u, \\ L_C &= \frac{1}{4}\partial_x^2 + \left(\frac{k_a}{2x} + \frac{a_u}{2}\right)\partial_x \\ &\quad + \frac{k_a(k_a-1)}{4x^2} + \frac{k_a a_u}{2x} + \frac{a'_u + a_u^2}{4}. \end{aligned} \quad (\text{CF26})$$

Substitute $f = x^{-k_a}u_a^{-1}g$ and use the product rule to obtain $L_C f = x^{-k_a}u_a^{-1}g''/4$. Alternatively, the first identity and $C''/C = c' + c^2$ give CF26 directly from CF18. At a negative even point the potential has coefficient $1/(2x^2)$; at one its x^{-2} coefficient is zero. All remaining singular and regular coefficients depend on the retained u_a and are present in CF26.

CF8. Fibre and jet maps through the exceptional points. The line-module isomorphism CF21 descends, for every integer $r \geq 1$, to

$$\mathcal{O}(D_C)_a / \mathfrak{m}_a^r \mathcal{O}(D_C)_a \xrightarrow[\mu_C]{\sim} \mathcal{O}_a / \mathfrak{m}_a^r, \quad [f] \mapsto [s(s-1)\pi^{-s/2}\Gamma(s/2)f]. \quad (\text{CF27})$$

It is well defined because μ_C and its inverse are $\mathcal{O}_a$ -linear and hence preserve $\mathfrak{m}_a^r$ multiples. These maps commute with all quotient maps from order $r+1$ to order r , as follows from the same formula. Thus they receive the entire local jet tower without truncating the unit in CF22.

For $r = 1$, these fibre maps are particularly concrete:

a	source fibre coordinate for f	μ_C on that coordinate
1	$\text{Res}_{s=1} f$	$\text{Res}_{s=1} f$
$-2n$	$f'(-2n)$	$\kappa_n f'(-2n)$
0	$f(0)$	$-2f(0)$.

(CF28)

At one, the source fibre is $x^{-1}\mathcal{O}_a/\mathcal{O}_a$; CF8 makes the multiplier on its residue coordinate one. At $-2n$, it is $x\mathcal{O}_a/x^2\mathcal{O}_a$, whose coefficient is $f'(a)$, and CF9 makes the multiplier κ_n . At zero, CF12 gives the factor -2 . These proofs identify the domains as well as the values. In particular the ordinary value of z_t at a negative even integer is zero, while its class in the twisted line fibre is nonzero and maps to the positive number $h_t(-2n)$.

Multiplication of two sections of $\mathcal{O}(D_C)$ belongs naturally to $\mathcal{O}(2D_C)$, not to a fixed untwisted ring of values. The complete multiplicative receiver is the graded isomorphism

$$\bigoplus_{j \geq 0} \mathcal{O}(jD_C) \longrightarrow \bigoplus_{j \geq 0} \mathcal{O}, \quad f_j \mapsto [s(s-1)\pi^{-s/2}\Gamma(s/2)]^j f_j. \quad (\text{CF29})$$

Indeed $C^{j+k}f_jg_k = (C^j f_j)(C^k g_k)$, proving compatibility with the graded product. No claim that the degree-one map is an untwisted ring homomorphism is needed or made.

CF9. Complete divisor relation and finite contour receivers. For every real t , h_t is not identically zero by CF5. Local orders in the exact product CF15 give

$$\text{div } h_t = \text{div } z_t + \text{div} \left(s(s-1)\pi^{-s/2}\Gamma(s/2) \right) = \text{div } z_t + [1] - \sum_{n \geq 1} [-2n]. \quad (\text{CF30})$$

There is no point-zero term at $s = 0$. The locally finite sums are legitimate because zeros and poles of a nonzero meromorphic function are discrete. At every nonexceptional point the full multiplier is a holomorphic unit, and CF15 therefore preserves the zero multiplicity exactly. At the exceptional real points CF16 evaluates the cancelling orders rather than presuming them.

Let Ω be a bounded domain with a finite, piecewise smooth boundary $\mathcal{C}$, oriented positively, and assume the boundary avoids all zeros and poles of h_t, z_t, C . If A is holomorphic on a neighborhood of $\bar{\Omega}$, define the signed trace of a meromorphic function g by

$$T_A(g; \mathcal{C}) = \frac{1}{2\pi i} \int_{\mathcal{C}} A(s) \frac{g'(s)}{g(s)} ds. \quad (\text{CF31})$$

Near a zero or pole, write $g = x^m v$ with v holomorphic and nonzero. Then $g'/g = m/x + v'/v$, so its weighted residue is $mA(a)$. The residue theorem proves that CF31 is the sum of the weighted zero orders, with pole orders negative. Applying the exact logarithmic derivative

$$\frac{h'_t}{h_t} = \frac{z'_t}{z_t} + \frac{\partial_s [s(s-1)\pi^{-s/2}\Gamma(s/2)]}{s(s-1)\pi^{-s/2}\Gamma(s/2)}$$

gives the finite receiver

$$T_A(h_t; \mathcal{C}) = T_A(z_t; \mathcal{C}) + \mathbf{1}_{\Omega}(1)A(1) - \sum_{\substack{n \geq 1 \\ -2n \in \Omega}} A(-2n). \quad (\text{CF32})$$

Only finitely many negative even points lie in the bounded domain. For a general finite closed cycle, replace each indicator by its signed winding number. This proves all endpoint signs with the original orientation.

If the fixed test A is itself meromorphic inside the boundary, the exact receiver is instead

$$T_A(h_t; \mathcal{C}) - T_A(z_t; \mathcal{C}) = \sum_{a \in \Omega} \text{Res}_{s=a} \left[A(s) \frac{\partial_s [s(s-1)\pi^{-s/2}\Gamma(s/2)]}{s(s-1)\pi^{-s/2}\Gamma(s/2)} \right]. \quad (\text{CF33})$$

The sum is finite and includes poles of A . To compute a residue even when a test pole meets an exceptional point, retain the actual local expansions

$$A(a+x) = \sum_{j=-q}^{\infty} A_j x^j, \quad \frac{C'}{C}(a+x) = \frac{k_a}{x} + \sum_{j \geq 0} b_j x^j.$$

Multiplication gives exactly

$$\text{Res}_a(Ac) = k_a A_0 + \sum_{j=1}^q A_{-j} b_{j-1}. \quad (\text{CF34})$$

This includes every finite-part contribution from the original full unit u_a . In particular, although $k_0 = 0$, a test $A(s) = 1/s$ receives the residue $c(0)$ in CF13, and a test $A(s) = 1/s^2$ receives $c'(0)$. Their constants are lost if the full multiplier is replaced by its divisor alone.

CF10. Full split support and retained factor presentations. First fix any bounded distributive support lattice L . For each of the complex vector spaces of germs or jets above, retain the synchronized carrier

$$M_L(V) = (V \times \{1_L\}) \cup (\{0\} \times L). \quad (\text{CF35})$$

Addition uses amplitude addition and support join; scalar multiplication by $(a, \nu) \in M_L(\mathbb{C})$ uses av and support meet $\nu \wedge \lambda$. The maps CF21 and CF27 have the exact lifts

$$\hat{\mu}_C(f, \lambda) = \left(s(s-1)\pi^{-s/2}\Gamma(s/2) f, \lambda \right), \quad \hat{\mu}_C^{-1}(h, \lambda) = (h/C, \lambda). \quad (\text{CF36})$$

They preserve every zero label and are mutually inverse on the stated line-module stalks and jet quotients. Proof consists of substituting CF21 in the carrier operations: linearity handles amplitudes, while the unchanged label makes each join and meet commute. At a negative even point, CF28 shows that the nonzero line-fibre amplitude maps to $h_t(-2n) \neq 0$; ordinary evaluation of the underlying function gives zero, a different explicit map whose kernel contains that line ideal. The fibre and evaluation maps are therefore both specified.

There is also an exact support-valued divisor receiver for the original factor presentation. It is useful to define its domain, since labels cannot be recovered from an unmarked meromorphic product. At a point a , take finite words of labelled nonzero meromorphic germs (f_j, λ_j) . The product map sends the word to $\prod_j f_j$. Retain the additive monoid

$$\mathbb{Z} \times L, \quad (m, \lambda) \oplus (n, \mu) = (m+n, \lambda \vee \mu).$$

The divisor receiver is

$$\nu_a^L((f_j, \lambda_j)_j) = \left(\sum_j \text{ord}_a f_j, \bigvee_{\{j: \text{ord}_a f_j \neq 0\}} \lambda_j \right). \quad (\text{CF37})$$

It is a homomorphism from concatenation of words to $\oplus$. Projection onto $\mathbb{Z}$ is exactly the ordinary valuation of the product, because orders add. This proves the connecting map, rather

than equating a factor word with its product. CF37 is an additive divisor receiver; it does not redefine multiplication in the supported scalar carrier CF35.

Apply it to the retained original word

$$(s, \lambda_s), (s-1, \lambda_{s-1}), (\pi^{-s/2}, \lambda_\pi), (\Gamma(s/2), \lambda_\Gamma), (z_t, \lambda_z). \quad (\text{CF38})$$

For real t , its orders at the three types of exceptional stalk are respectively

a	s	$s-1$	$\pi^{-s/2}$	$\Gamma(s/2)$	z_t	sum
0	1	0	0	-1	0	0
1	0	1	0	0	-1	0
$-2n$	0	0	0	-1	1	0.

(CF39)

Thus the resulting amplitudes are zero, while their divisor support labels are, respectively, $\lambda_s \vee \lambda_\Gamma$, $\lambda_{s-1} \vee \lambda_z$, and $\lambda_\Gamma \vee \lambda_z$. Whenever those labels are nonzero in L , the corresponding supported zero differs from $(0, 0_L)$. Opposite orders therefore cancel in the ordinary divisor map without deleting the displayed factor labels. The complete units remain in the word CF38 and the exact map CF36; a divisor receiver alone does not encode their Taylor coefficients. This distinguishes both retained data and both proved projection maps.

CF11. The precise information lost by suppressing the factor. With $C = s(s-1)\pi^{-s/2}\Gamma(s/2)$ retained, CF21 is an exact equivalence of the line-module sections, CF25 is an exact intertwining of their differential operators, and CF30–34 are exact divisor and trace receivers. Thus passage to h_t is not a mathematical error. Identifying it with z_t after dropping these maps, however, loses specified information.

First, the actual original function $z_0 = \zeta$ and every z_t have the simple pole and simple real zeros in CF16. Their completed partner has none of those real zeros or poles by CF5. The relation is the signed divisor CF30, not equality of the two divisors. At zero their values differ by the exact factor -2 , and their first two multiplier derivatives are CF12–13.

Second, ordinary heat on z_t is false as an identity of meromorphic functions for every real t . Write near one $z_t(s) = a_t/(s-1) + O(1)$, where $a_t = 16H_t(i) > 0$. Because $z_t = C^{-1}h_t$ with fixed C , $\partial_t z_t$ has at most a simple pole there. But $z_t''/4$ has the nonzero triple-pole coefficient $a_t/2$. Hence

$$\partial_t z_t \neq \frac{1}{4} z_t''. \quad (\text{CF40})$$

The exact missing term is the specified differential operator

$$(L_C - \frac{1}{4}\partial_s^2)f = \frac{\partial_s[s(s-1)\pi^{-s/2}\Gamma(s/2)]}{2s(s-1)\pi^{-s/2}\Gamma(s/2)}f' + \frac{\partial_s^2[s(s-1)\pi^{-s/2}\Gamma(s/2)]}{4s(s-1)\pi^{-s/2}\Gamma(s/2)}f. \quad (\text{CF41})$$

CF25–26 prove its exact action through the exceptional stalks. The triple-pole contradiction is resolved by these retained terms, not by a change of time.

Third, even retaining only the divisor and the nonzero value at zero cannot reconstruct the unit data. To prove this failure of that particular forgetful map, consider, solely as a comparison,

$$C_{a,b}(s) = e^{as+bs^2}s(s-1)\pi^{-s/2}\Gamma(s/2). \quad (\text{CF42})$$

This does not replace the original C or define the transported zeta family. It has the same divisor and the same value -2 at zero for all complex a, b , but

$$\frac{C'_{a,b}}{C_{a,b}} = c + a + 2bs, \quad c_{a,b}(0) = c(0) + a, \quad c'_{a,b}(0) = c'(0) + 2b. \quad (\text{CF43})$$

Therefore the drift, potential, and meromorphic-test residues in CF18–19 and CF34 are not determined by those reduced data. The full original product CF2, its local units CF22, and its line and support maps CF21, CF27, CF36–39 recover exactly what is missing. All statements concern the original zeta function and its specified transported family. No assertion of an RH proof, disproof, or failure of the completed xi function is made.

3 Original-zeta divisor tests and their complete matrix receivers

OZ1. Original function, fixed coordinate, and reflection. The initial function is the meromorphic Riemann zeta function $\zeta(s) = \sum_{n \geq 1} n^{-s}$ for $\Re s > 1$, continued meromorphically. Retain the precise original Rodgers–Tao source [The de Bruijn–Newman constant is non-negative](#), equations `hoz`, `sas`, `phidef`, and `htdef`, and all four factors

$$C(s) = s(s-1)\pi^{-s/2}\Gamma(s/2), \quad Z(s) = -2i(s - \tfrac{1}{2}), \quad z_t(s) = C(s)^{-1}16H_t(Z(s)). \quad (\text{OZ1})$$

CF1–29 prove $z_0 = \zeta$, the full meromorphic heat operator, and its isomorphism of line modules with the auxiliary theta family. In particular neither C nor any derivative of it is discarded. For meromorphic tests put

$$F^\#(s) = \overline{F(1-\bar{s})}, \quad F_j(s) = Z(s)^{-(2j+1)}, \quad j \geq 0. \quad (\text{OZ2})$$

The identity $Z(1-\bar{s}) = \overline{Z(s)}$ proves $F_j^\# = F_j$. A test $F_c = \sum c_j F_j$ therefore has $F_c^\# = \sum \bar{c}_j F_j$. These are the actual rational functions in the original arithmetic coordinate; no coordinate rescaling is made. Their only finite pole is $s = 1/2$.

OZ2. All zeros and poles in the full signed divisor. For real t , CF3–17 prove that $16H_t(Z(s))$ is positive on the real s -axis. The full signed divisor of z_t is therefore exactly

$$\text{div } z_t = \sum_{H_t(\lambda)=0} m_\lambda [\tfrac{1}{2} + i\lambda/2] + \sum_{\ell \geq 1} [-2\ell] - [1]. \quad (\text{OZ3})$$

There is no duplication: the first sum has no real s , and CF16 proves that every displayed real zero and the pole is simple. In particular $s = 1/2$ is absent from this divisor, so no test pole meets a zero or pole of z_t .

Define, on these rational tests, the original full divisor pairing

$$\begin{aligned} B_t^\zeta(F_c, F_d) := & \sum_{H_t(\lambda)=0} m_\lambda F_c^\#(\tfrac{1}{2} + i\lambda/2) F_d(\tfrac{1}{2} + i\lambda/2) \\ & + \sum_{\ell \geq 1} F_c^\#(-2\ell) F_d(-2\ell) - F_c^\#(1) F_d(1). \end{aligned} \quad (\text{OZ4})$$

At $t = 0$ this is the full signed divisor of the original ζ . The notation at other times means the explicitly specified z_t , not a claim that $z_t = \zeta$. All sums converge absolutely. Indeed WR1–2 prove $\sum m_\lambda |\lambda|^{-2} < \infty$, and each product is $O(|\lambda|^{-2})$. The remaining series is bounded by a constant times $\sum (4\ell+1)^{-2}$. The finitely many small denominators are nonzero because $H_t(0) > 0$. These convergence statements justify all finite changes of basis and regroupings below.

OZ3. Exact arithmetic correction at every matrix size. Write the original complete auxiliary moments and the additional real-zero series as

$$S_n(t) = \sum_{H_t(\lambda)=0} m_\lambda \lambda^{-2n}, \quad T_n = \sum_{\ell=1}^{\infty} (4\ell+1)^{-2n}, \quad n \geq 1. \quad (\text{OZ5})$$

Both signs and the single copy of each trivial zero are essential. Since $Z(-2\ell) = i(4\ell+1)$ and $Z(1) = -i$, direct substitution in OZ4 gives

$$\begin{aligned} K_{ij}^\zeta(t) &= S_{i+j+1}(t) + (-1)^{i+j+1} (T_{i+j+1} - 1), \\ K_{ij}^C &= (-1)^{i+j+1} (1 - T_{i+j+1}), \\ K_d^h(t) &= K_d^\zeta(t) + K_d^C, \quad B_t^\zeta(F_c, F_d) = c^* K_d^\zeta(t) d. \end{aligned} \quad (\text{OZ6})$$

Here $K_{ij}^h = S_{i+j+1}$. Thus OZ6 proves the exact matrix map at every finite rank, including the previous ranks 19, 63 and 64. The matrix K^C is the signed divisor contribution of the actual full multiplier C , by CF10. All three displayed matrices are real symmetric. Consequently OZ4 is Hermitian on this specified rational-test space. No Hermiticity assertion is made for arbitrary meromorphic tests on the nonsymmetric full zeta divisor.

OZ4. A complete time-zero value with every trivial zero. Take the fixed original test

$$F(s) = F_0(s) + F_1(s) = Z(s)^{-1} + Z(s)^{-3}. \quad (\text{OZ7})$$

At the pole $s = 1$, its value is zero. This is a present zero of an explicitly evaluated test, not deletion of the pole from OZ3. At -2ℓ , put $a_\ell = (4\ell + 1)^{-2}$. The value of F is $-i\sqrt{a_\ell}(1 - a_\ell)$. As $F^\# = F$, its divisor contribution is exactly $-a_\ell(1 - a_\ell)^2$. Therefore

$$\begin{aligned} B_0^\zeta(F, F) &= S_1(0) + 2S_2(0) + S_3(0) - (T_1 - 2T_2 + T_3), \\ B^C(F, F) &= T_1 - 2T_2 + T_3 > 0, \\ B_0^\zeta(F, F) + B^C(F, F) &= S_1(0) + 2S_2(0) + S_3(0). \end{aligned} \quad (\text{OZ8})$$

The first trivial zero alone contributes $-576/15625$. Every further trivial-zero contribution has the same strict negative sign. The original complete-theta interval calculation described below gives

$$0.0115617942 < S_1(0) + 2S_2(0) + S_3(0) < 0.0115617943 < 3/250. \quad (\text{OZ9})$$

It follows already from the first trivial zero that $B_0^\zeta(F, F) < 3/250 - 576/15625 < 0$. Including all remaining trivial zeros gives the certified wider decimal enclosures

$$\begin{aligned} -0.059716 &< B_0^\zeta(F, F) < -0.059714, \\ 0.071276 &< B^C(F, F) < 0.071278. \end{aligned} \quad (\text{OZ10})$$

Thus the negative sign here comes with an explicitly computed positive compensating term; it is not a negative value of the nontrivial-zero Weil pairing. No RH disproof follows from OZ10.

OZ5. Reproducible bounds and the exact numerical inputs. The complete-theta computation used here is the retained RPX3 calculation `reproduce_witness63.py`, whose full binary interval receipt has SHA256 `aa296253ed03decc4bf11da1d2be897ab5eef990c65370c922e1abcc8d4e3b56`. Its time-zero input uses the full Rodgers–Tao kernel, not a list of zeros. The analytic quadrature proof is [WR4–5](#), [equations WR8–11](#); RPX3 retains 24 theta summands, 300 bracketed Gauss nodes on each of 24 panels of $[0, 3]$, 900-digit directed intervals, and the two explicit tails

$$192\,25^4 3^{2m} e^{27-1875}, \quad e^{-2000} \quad (0 \leq m \leq 129).$$

The interval library is not asserted to be formally verified. The accompanying exact-arithmetic checker reads its binary endpoints without rounding. For clarity the moment receiving map is also given here. With

$$M_n = \int_0^\infty u^{2n} \Phi(u) du, \quad a_n = (-1)^n M_n / (2n)!,$$

the even Hadamard product, justified by the WR1 growth and absolute reciprocal-square bound, gives $H'_0/H_0 = -\sum_{n \geq 1} S_n Z^{2n-1}$. Multiplication by $H_0 = \sum a_n Z^{2n}$ yields exactly

$$S_n = \frac{-2na_n - \sum_{j=1}^{n-1} a_j S_{n-j}}{M_0}. \quad (\text{OZ11})$$

The positive original M_0 remains in this map. The receipt stores a dummy $S_0 = 0$ at index zero, so its entries with indices 1, 2, 3 are exactly those in OZ9.

The additional series are bounded with rational arithmetic alone. For $N = 400$, monotonicity of $(4x + 1)^{-2n}$ proves

$$\sum_{\ell=1}^N (4\ell + 1)^{-2n} + \frac{1}{4(2n-1)(4(N+1)+1)^{2n-1}} \leq T_n \quad (\text{OZ12})$$

$$\leq \sum_{\ell=1}^N (4\ell + 1)^{-2n} + \frac{1}{4(2n-1)(4N+1)^{2n-1}}.$$

Indeed the sum of the decreasing function after N lies between its integrals from $N+1$ and from N , and direct differentiation gives the displayed antiderivative. No trivial zero is omitted. Applying rational interval arithmetic to OZ6 and OZ8 gives OZ9–10, and also

$$K_{00}^\zeta(0) > 0, \quad -0.059715 < \det K_2^\zeta(0) < -0.059713. \quad (\text{OZ13})$$

A real symmetric two-by-two matrix with negative determinant has one positive and one negative eigenvalue: its real eigenvalues have negative product and neither is zero. Hence the original full divisor matrix has inertia $(1, 1, 0)$, proved on the same two original functions. The complete arithmetic intervals, source pin and checker are `FULL_ZETA_TRACE_RECEIPT.json` and `evaluate_full_zeta_trace.py`.

OZ6. Derivatives, coefficient maps, and every earlier compression. On every real time interval all the moments are analytic by CF3–5 and WR4. The additional matrix in OZ6 is constant in the original physical time. Thus, for every positive integer r ,

$$\partial_t^r K_d^\zeta(t) = \partial_t^r K_d^h(t). \quad (\text{OZ14})$$

For an arbitrary specified, fixed coefficient matrix J , including the earlier exact conditioning diagonal and its nineteen-to-two coefficient injection, the actual receiving identity is

$$G^\zeta = J^* K^\zeta J, \quad D_J = J^* K^C J, \quad G^h = G^\zeta + D_J. \quad (\text{OZ15})$$

This follows by multiplying OZ6 on both sides; it keeps the original coefficient map. Neither equality of inertias nor equality of singular times is asserted. In particular the old auxiliary matrix is not renamed the original full-divisor matrix.

Here is the exact connection correction, not merely an instruction to recalculate it. On the actual open set where both matrices in OZ15 are invertible, set $\Gamma_\zeta = \frac{1}{2}(G^\zeta)^{-1}(G^\zeta)'$ and $\Gamma_h = \frac{1}{2}(G^h)^{-1}(G^h)'$. The resolvent identity gives

$$\Gamma_\zeta - \Gamma_h = \frac{1}{2}(G^\zeta)^{-1} D_J (G^h)^{-1} (G^\zeta)'. \quad (\text{OZ16})$$

To verify it, multiply $(G^\zeta)^{-1} - (G^h)^{-1} = (G^\zeta)^{-1}(G^h - G^\zeta)(G^h)^{-1}$ by the common derivative in OZ14. This specifies the exact domain and every term, without an assumption of equality of connections. For a fixed path in that open set, the original fundamental solutions defined by $\partial_t P = -\Gamma P$, $P(s, s) = I$, satisfy

$$P_\zeta(t, s) - P_h(t, s) = - \int_s^t P_\zeta(t, v) (\Gamma_\zeta(v) - \Gamma_h(v)) P_h(v, s) dv. \quad (\text{OZ17})$$

Differentiate $P_\zeta(t, v) P_h(v, s)$ with respect to v and integrate. For a complex path use the pathwise differential and its orientation. The same identity then includes all transport along a loop. A previously computed reflection cannot be assigned to this changed matrix without this correction.

For the attained Schur map, retain a specified block decomposition $G = \begin{pmatrix} A & B \\ C & E \end{pmatrix}$. The exact source inclusion $x \mapsto (x, -E^{-1}Cx)$ gives $A - BE^{-1}C$ whenever the original block E is invertible and G is Hermitian. Applying this same calculation to $G^h - D_J$ gives the full original-divisor receiver

$$A_h - D_{11} - (B_h - D_{12})(E_h - D_{22})^{-1}(C_h - D_{21}). \quad (\text{OZ18})$$

All four blocks of the correction are present. The expression is defined precisely on the open set where its displayed denominator is invertible; it is not a rank-based allocation. Direct block elimination also gives $\det G^\zeta = \det(E_h - D_{22}) \det(\text{OZ18})$. This proves the determinant receiver for every prior block choice.

OZ7. Full support and finite contour interpretation. The exact lift of every linear coefficient map in OZ15 is the full $G_L(\mathbb{C})$ -module map of CF35–36, with all $\lambda \in L$ retained. Finite contractions are performed with the original supports. In particular the two nonzero terms $F_0(1)$ and $F_1(1)$ in OZ7 cancel to supported $e = (0, 1_L)$; they do not produce $\tau = (0, 0_L)$. The divisor with its listed pole and the zero test value are both retained. Infinite amplitude sums in OZ4 carry the explicitly specified full support of this test construction; no nonexistent arbitrary join in a merely bounded lattice is invoked.

The divisor sums above are not silently equated with uncorrected contour integrals. The meromorphic test $A = F_c^\# F_d$ has a pole at $s = 1/2$. On a bounded positive contour avoiding all zeros and poles, the identity between the three logarithmic derivatives is exact. If its interior contains the test pole, the residue of AC'/C at that pole must also be kept. Write $A = \sum A_j(s - 1/2)^j$ and $C'/C = \sum_{j \geq 0} b_j(s - 1/2)^j$ there. The additional residue is exactly $\sum_{j \geq 1} A_{-j}b_{j-1}$. The analogous regular logarithmic-derivative residues occur for h_t and z_t . CF33–34 prove the full contour identity including all of them. OZ4–18 use absolutely convergent divisor sums directly and never discard these contour contributions.

OZ8. The actual earlier two-test matrix after reconstruction. Retain exactly the supplied nineteen integers $n_0, \dots, n_{18}$ and denominator 10^{120} in `rational_witness.json`, with source SHA256 `85314e9fb3115647b7f03a5e00f1b5512c4b23c9280a753f3cf2cababdfed4ac`. Their original test and its unchanged companion are

$$f_*(s) = \sum_{i=0}^{18} c_i F_i(s), \quad c_i = 1000^i n_i / 10^{120}, \quad F_0(s) = Z(s)^{-1}. \quad (\text{OZ19})$$

The complete integers remain in the accompanying source; no rounded coefficient is used. Let Q_h, B_h, A_h denote the original complete-theta contractions in this exact ordered pair (f_*, F_0) , as in NF2. With $E_n = (-1)^n(T_n - 1)$, OZ6 gives, term by term,

$$\begin{aligned} Q_\zeta &= Q_h + \sum_{i,j=0}^{18} c_i c_j E_{i+j+1}, \\ B_\zeta &= B_h + \sum_{i=0}^{18} c_i E_{i+1}, \\ A_\zeta &= A_h + E_1, \quad G_\zeta = \begin{pmatrix} Q_\zeta & B_\zeta \\ B_\zeta & A_\zeta \end{pmatrix}. \end{aligned} \quad (\text{OZ20})$$

These are the actual original-zeta values of the earlier tests, not newly chosen tests adapted to a desired sign.

For the whole original real interval $I = [-2, -2 + 10^{-20}]$, the retained `NATIVE_SCHUR_INTERVALS.json` encloses the three original theta contractions with all summands and spatial tails. Its proof is NF2, NC6–10, and WR4–5: the whole-window integral uses e^{tu^2} , $t \in I$, inside the

directed interval calculation, followed by the exact recurrence OZ11. Thus it encloses every time in the interval, not only the endpoints. The new checker `propagate_native_original_zeta.py` reads those binary intervals, applies OZ12 for every $1 \leq n \leq 37$, and substitutes OZ20 at 400-digit directed precision. Its full receipt `NATIVE_ORIGINAL_ZETA_RECEIPT.json` proves

$$\begin{aligned} A_\zeta(t) &> 0, & Q_\zeta(t) &> 0, \\ -6.41746 \cdot 10^{106} &< \det G_\zeta(t) < -6.41741 \cdot 10^{106}, \\ -6.85437 \cdot 10^{106} &< Q_\zeta(t) - B_\zeta(t)^2/A_\zeta(t) < -6.85431 \cdot 10^{106}, & t \in I. \end{aligned} \tag{OZ21}$$

The large scale is a consequence of the unchanged original coefficients and the retained real divisor evaluations. No scale factor has been removed. OZ21 follows by interval multiplication, subtraction and division with the strictly positive denominator A_ζ . The integral-tail proof OZ12 supplies every omitted trivial-zero term. The previous quadrature interval library remains a computer-assisted input, not a formally verified library.

The determinant sign proves that the full original-zeta matrix has inertia $(1, 1, 0)$ throughout this interval. In particular it has no singular time there. Its actual attained negative test is the explicit coefficient section

$$\begin{aligned} f_*(s) - \frac{B_\zeta(t)}{A_\zeta(t)} F_0(s), \\ B_t^\zeta \left(f_* - \frac{B_\zeta}{A_\zeta} F_0, f_* - \frac{B_\zeta}{A_\zeta} F_0 \right) = Q_\zeta - \frac{B_\zeta^2}{A_\zeta} < 0. \end{aligned} \tag{OZ22}$$

Expansion of the two-by-two form proves the equality, with the original real coefficients. Its original-zeta connection has no pole at any point of I . To prove this local complex statement, the entries have their unique holomorphic continuation near each real time since $H_t(0) > 0$. The nonzero determinant gives a complex disk about that point on which the inverse is holomorphic. Thus

$$\Gamma_\zeta = \frac{1}{2} G_\zeta^{-1} G'_\zeta \quad \text{is holomorphic locally at each } t \in I. \tag{OZ23}$$

On a simply connected sufficiently small disk the holomorphic linear equation has a single-valued invertible fundamental solution (successive integration of its convergent analytic coefficients constructs it). Therefore its local loop transport is identity. The compressed reflection of the earlier G_h at its own singular time is not the local monodromy of this new matrix. OZ16–17 give the exact connecting transport away from that old singularity. All these signs concern the uncompensated full divisor, with the compensating matrix still specified by OZ6.

OZ9. What has actually been reconstructed. The original meromorphic function, its pole, all trivial zeros, the full multiplier, every finite coefficient map, the connection change, and the original Schur blocks are explicit in OZ1–18. The earlier theta-source estimates still calculate their stated S_n ; their full original-zeta receiver is now OZ6, rather than an identification of the two divisors. Positivity of one form is not inserted as an assumption about the other. The evaluated time-zero difference OZ8–13 is evidence of the exact contribution restored here, with no assertion that RH was assumed by the faithful line-module equivalence CF21.

4 A faithful labelled heat source and the original theta endpoints

LH1. The received obstruction specifies a construction. The supplied continuation *Untitled 1793* proposes the shifted Gaussian sum with its zero mode retained. We retain that proposal and prove its domains, its source maps and an exact return to the original zeta function. No assertion that every heat operation is already an endomorphism of the arithmetic semiring is

used. Let L be any nontrivial bounded distributive lattice, with no finiteness assumption, and retain

$$S = G_L(\mathbb{Z}) = \{(n, 1_L) : n \in \mathbb{Z}\} \cup \{(0, \lambda) : \lambda \in L\}, \quad \tau = (0, 0_L), \quad e = (0, 1_L). \quad (\text{LH1})$$

There is one copy of $(0, 1_L)$. The coefficient operations are $(a, \lambda) + (b, \mu) = (a + b, \lambda \vee \mu)$ and $(a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu)$. Carrier membership holds because a nontop label forces zero amplitude; distributivity follows from ring distributivity and the distributive lattice law. The supported-zero ideal is prime: it is precisely $\{(0, \lambda) : \lambda \in L\}$, since multiplying e by these same elements gives all of them. A product belongs to it exactly when its integer amplitudes have zero product, hence one amplitude is zero. It is proper because it excludes $(1, 1_L)$. This proves the required prime directly without deleting any label.

The phrase that both zero elements must map to zero under every multiplicative observation is too strong. The amplitude map $p : S \rightarrow \mathbb{R}$, $p(n, 1) = n$, $p(0, \lambda) = 0$, is a unital semiring map and does identify the whole zero fibre. But every unital multiplicative map to $\mathbb{R}$ sends e to an idempotent, hence either zero or one. If it sends e to one, the equations $e(n, 1) = e$ force the image of every supported integer to be one. This case occurs: the support character of $G_{\{0,1\}}(\mathbb{Z})$ sends all supported integers to one and τ to zero, multiplicatively. It is not additive into the real ring, since $1 + (-1) = e$ would give $2 = 1$. For every additive map to a characteristic-zero vector space, instead, $z_\lambda + z_\lambda = z_\lambda$ forces the image of z_λ to zero. These are exact restrictions on the respective maps, not an irrelevance argument about the zero fibre.

LH2. A source which keeps every labelled point. Let $V_L = \bigoplus_{\lambda \in L} \mathbb{C}v_\lambda$, with independent basis vectors, and let $\mathcal{D}'_L$ be the direct sum of tempered-distribution spaces indexed by L . Finite linear combinations of the following embedded points form a subspace $\mathcal{A}_L$:

$$\iota(n, 1_L) = \delta_n v_{1_L}, \quad \iota(0, \lambda) = \delta_0 v_\lambda \quad (\lambda \neq 1_L). \quad (\text{LH2})$$

This is an injective map of sets, and its linear extension is an isomorphism from the free complex vector space on the set S to $\mathcal{A}_L$. To prove independence, separate the lattice coordinates. In the top coordinate the finitely many Dirac masses are independent: smooth compactly supported functions in disjoint small intervals around their integers extract each coefficient. The other coordinates each contain one independent Dirac mass. This free construction retains the original coefficient operations as the bilinear point maps

$$\begin{aligned} m_+(\delta_a v_\lambda \otimes \delta_b v_\mu) &= \delta_{a+b} v_{\lambda \vee \mu}, \\ m_\times(\delta_a v_\lambda \otimes \delta_b v_\mu) &= \delta_{ab} v_{\lambda \wedge \mu}. \end{aligned} \quad (\text{LH3})$$

They are defined on the displayed source bases, extended linearly on $\mathcal{A}_L \otimes \mathcal{A}_L$. On the original embedded points they reproduce every original operation, including the additive identity τ , the multiplicative unit and both zero elements. We do not assert that ordinary vector addition reproduces the original semiring addition; the proved map is LH3. For example $\iota(e)$ is a nonzero vector although $e + e = e$.

LH3. Positive-time heat with an exact inverse on its image. For actual heat time $T > 0$ define

$$k_T(x) = (4\pi T)^{-1/2} \exp[-x^2/(4T)], \quad \mathcal{H}_T U = k_T * U \quad (\text{LH4})$$

coordinatewise. Differentiation gives $\partial_T k_T = \partial_x^2 k_T$, including the derivative of the complete prefactor. The Gaussian Fourier transform, in the original convention $\hat{f}(\xi) = \int f(x) e^{-2\pi i x \xi} dx$, is

$$\hat{k}_T(\xi) = e^{-4\pi^2 T \xi^2}. \quad (\text{LH5})$$

For a direct proof, integration by parts yields $I'(\xi) = -8\pi^2 T \xi I(\xi)$; the Gaussian integral gives $I(0) = 1$. Solving this first-order equation proves LH5. Completing the square in the absolutely convergent convolution integral, retaining both prefactors, gives $k_T * k_U = k_{T+U}$.

Convolution by this Schwartz function is defined on each tempered distribution and has Fourier multiplier LH5. It is injective. Indeed if $e^{-4\pi^2 T \xi^2} \widehat{U} = 0$, test locally against a compactly supported smooth function ϕ , after dividing that test by the smooth nonzero multiplier. This gives $\langle \widehat{U}, \phi \rangle = 0$. Compact-test uniqueness implies $\widehat{U} = 0$ and hence $U = 0$. Thus the inverse on the actual range is the distributional operation

$$U = \mathcal{F}^{-1} \left(e^{4\pi^2 T \xi^2} \widehat{\mathcal{H}_T U} \right). \quad (\text{LH6})$$

This inverse is asserted on this range; multiplying an arbitrary tempered distribution by the growing exponential need not yield a tempered distribution. No unrestricted backward heat assertion is made. Every finite support label is retained separately.

In particular $\mathcal{H}_T \iota(e) = k_T v_{1_L}$ and $\mathcal{H}_T \iota(\tau) = k_T v_{0_L}$ are different nonzero vectors for every positive time. The exact original arithmetic operations on their heat images are

$$m_{+,T} = \mathcal{H}_T m_+ (\mathcal{H}_T^{-1} \otimes \mathcal{H}_T^{-1}), \quad m_{\times,T} = \mathcal{H}_T m_{\times} (\mathcal{H}_T^{-1} \otimes \mathcal{H}_T^{-1}). \quad (\text{LH7})$$

Their domain is $\mathcal{H}_T \mathcal{A}_L \otimes \mathcal{H}_T \mathcal{A}_L$, and their output is $\mathcal{H}_T \mathcal{A}_L$. Substitution gives, for example, $m_{+,T}(k_T(x-a)v_\lambda, k_T(x-b)v_\mu) = k_T(x-a-b)v_{\lambda \vee \mu}$, and the product receiver has ab and $\lambda \wedge \mu$. This supplies explicit faithful heat images of the semiring points and their operations. It does not identify these operations with pointwise addition and multiplication of heat functions.

LH4. The actual integer comb and its marked zero. The infinite integer comb $D = \sum_{n \in \mathbb{Z}} \delta_n$ is tempered, because every Schwartz function has summable values on the integers. Split its source as the retained pair $(D - \delta_0, \delta_0)$, whose sum is D . With a finite set of lower labels $J \subseteq L \setminus \{1_L\}$, retain the source

$$\mathbf{D}_J = D v_{1_L} + \delta_0 \sum_{\lambda \in J} v_\lambda. \quad (\text{LH8})$$

No infinite sum of unrelated lattice coordinates is needed. The directed union over finite J retains every individual lower label. Heat gives exactly

$$\mathcal{H}_T \mathbf{D}_J(x) = \sum_{n \in \mathbb{Z}} k_T(x-n) v_{1_L} + k_T(x) \sum_{\lambda \in J} v_\lambda. \quad (\text{LH9})$$

The separate marked top zero term is $k_T(x) v_{1_L}$. It satisfies the same heat equation and has mass one for all $T > 0$. Thus this construction moves the marked zero profile without erasing it. Applying the inverse LH6 componentwise to the retained pair recovers that pair and every lower coordinate. The sum alone does not specify an arbitrary choice of marked summand.

The top coordinate in LH9 is periodic, but the separate marked Gaussian is not periodic. Periodizing that separate Gaussian gives the top circle heat function and thereby forgets which integer was originally marked. Keeping the pair avoids this additional loss. This identifies a precise limitation of scalar circle periodization while retaining its connecting map.

LH5. Shifted theta and its two arithmetic sheets. Set $x = (4\pi T)^{-1}$ and use a separate displacement variable $b \in \mathbb{R}$. Then the exact top heat kernel is

$$\sum_{n \in \mathbb{Z}} k_T(b-n) = \sqrt{x} K(x; b), \quad K(x; b) = \sum_{n \in \mathbb{Z}} e^{-\pi(n+b)^2 x}. \quad (\text{LH10})$$

Changing n to $-n$ proves the sign in this equality. Direct differentiation or LH4 proves the heat equation in (T, b) ; b is displacement, not the Rodgers–Tao time. On compact $x > 0$ ranges

every derivative converges by a Gaussian majorant. The original marked top zero and each lower coordinate have the term $E(x; b) = e^{-\pi b^2 x}$. The exact labelled comparison is

$$((K - E)v_{1_L}, Ev_{1_L}, (Ev_\lambda)_{\lambda \in J}), \quad (\text{LH11})$$

with the specified sum map to LH9 after multiplication by $\sqrt{x}$. All components remain available before that sum.

For $0 < b < 1$, $\Re s > 1$, termwise Mellin integration yields

$$\int_0^\infty K(x; b)x^{s/2-1}dx = \pi^{-s/2}\Gamma(s/2) \left[b^{-s} + \sum_{n \geq 1} (n+b)^{-s} + \sum_{n \geq 1} (n-b)^{-s} \right]. \quad (\text{LH12})$$

To prove interchange, replace s by its real part in the absolute value, integrate each positive Gaussian by $y = \pi(n+b)^2x$, and use convergence of the resulting inverse-power series. The two last sums are exactly $\zeta(s, 1+b)$ and $\zeta(s, 1-b)$ on this half-plane. Their named functions are not substitutes for either full series or the displayed marked term b^{-s} .

At $b = 0$ the marked term is exactly one as a function of x , and its integral over $(0, \infty)$ has no common convergence half-plane. The correct retained pair has the unmarked Mellin identity

$$\int_0^\infty (K(x; 0) - 1)x^{s/2-1}dx = 2\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \Re s > 1, \quad (\text{LH13})$$

alongside the separate marked constant and all lower labels. Consequently it recovers the original ζ by multiplying by $\pi^{s/2}/[2\Gamma(s/2)]$. This is an equality with every factor retained, not an identification of the two functions.

LH6. Both endpoint terms, derived from the actual Gaussian. For completeness the Gaussian Poisson identity needed here follows without a formal deletion of the zero term. The smooth periodic sum $K(x; b)$ has Fourier coefficient at integer m equal to $x^{-1/2}e^{-\pi m^2/x}$: unfold the sum over a unit interval and apply the Gaussian Fourier calculation LH5. Its absolutely convergent Fourier series therefore gives

$$K(x; b) = x^{-1/2} \sum_{m \in \mathbb{Z}} e^{-\pi m^2/x} e^{2\pi i m b}. \quad (\text{LH14})$$

At zero displacement set $\psi(x) = \sum_{n \geq 1} e^{-\pi n^2 x}$. The zero Fourier coefficient and the zero source coefficient give separately $\psi(x) = x^{-1/2}\psi(1/x) + (x^{-1/2} - 1)/2$. Split the Mellin integral of ψ at one and substitute $y = 1/x$ only on $(0, 1)$. Direct integration of each of the two endpoint terms then proves

$$\zeta(s) = \frac{\pi^{s/2}}{\Gamma(s/2)} \left[\frac{1}{s-1} - \frac{1}{s} + \int_1^\infty \psi(x) \left(x^{s/2-1} + x^{-(s+1)/2} \right) dx \right]. \quad (\text{LH15})$$

The integral is entire in s , since the Gaussian decay on $[1, \infty)$ controls all powers of $\log x$ on compact sets. Thus LH15 extends meromorphically and retains both endpoint terms. It agrees with the original-source theta formula used by [Rodgers and Tao](#), while CF1–29 supply the exact subsequent differential and line-module map. The supplied continuation's proposed shifted sum is thereby proved as an exact scalar Mellin representation, with the faithful labelled distribution source retained by LH11. The scalar shifted sum itself has $K(x; b) = K(x; 1-b)$ for $0 < b < 1$, by the integer change of index, so it does not determine the displacement on that whole interval. Keeping $E(x; b) = e^{-\pi b^2 x}$ resolves this ambiguity there: for any fixed $x > 0$ this is strictly decreasing in $b > 0$, with inverse $b = \sqrt{-\log E(x; b)/(\pi x)}$.

LH7. The obstruction and its computed consequence. The actual lower labels do not enter the scalar series in LH15: projection onto the top coordinate has kernel precisely the direct sum of the lower-coordinate distribution spaces. Its restriction to the linear span of Dv_{1_L} and $\{\delta_0 v_\lambda : \lambda \in J\}$ has the finite kernel spanned by $\delta_0 v_\lambda$, $\lambda \in J$, whose heat images are the nonzero $k_T v_\lambda$. Keeping LH9 and LH11 makes the source and its heat evolution faithful; projecting loses exactly that identified subspace. The labelled heat generator acts there by ∂_x^2 , which is nonzero on its Gaussian images. Thus this is an explicit alternative positive-time heat construction with an active marked zero mode. No assertion that its positivity or its ordinary circle spectrum settles the original RH is part of the calculation. The maps LH2–15 specify which original data it retains and how its scalar part returns to the original zeta.

5 The supported-zero action forces a retained mass coordinate

MCB1. Actual coefficient domain and arithmetic action. Fix a finite nontrivial bounded distributive lattice L . Its finiteness is used for the bounded lower-coordinate sums below; the prime and set-level construction LH1–15 retains arbitrary L . Let

$$\mathcal{V}_0 = c_{00}(\mathbb{Z}) \oplus \mathbb{C}^{L \setminus \{1_L\}}, \quad v = (a, b), \quad m(a) = \sum_{n \in \mathbb{Z}} a_n. \quad (\text{MCB1})$$

The coefficients are those of the free original-point embedding LH2: a_n is at $(n, 1_L)$ and b_λ at $(0, \lambda)$. In particular a_0 records supported zero. Multiplication of these original semiring points by $e = (0, 1_L)$ gives the exact linear source operator

$$E(a, b) = (m(a)\delta_0, b), \quad E^2 = E. \quad (\text{MCB2})$$

Indeed $e(n, 1_L) = e$ for every integer n , and $e(0, \lambda) = (0, \lambda)$. This determines all columns of the operator and proves idempotence, since $m(m(a)\delta_0) = m(a)$. It is a free-source action of the original point multiplication, not a new assignment to supported zero.

MCB2. The exact failure of the original coefficient Hilbert domain. Give $\mathcal{V}_0$ its coefficient norm $\|(a, b)\|_0^2 = \sum |a_n|^2 + \sum |b_\lambda|^2$. Let $a_n^{(N)} = 1/N$ for $1 \leq n \leq N$ and zero otherwise. Then

$$\|a^{(N)}\|_2^2 = 1/N \longrightarrow 0, \quad m(a^{(N)}) = 1, \quad E(a^{(N)}, 0) = (\delta_0, 0). \quad (\text{MCB3})$$

Thus E has no bounded extension to the coefficient Hilbert space. It is not even closable there: closability requires that $v_N \rightarrow 0$ and $Ev_N \rightarrow w$ imply $w = 0$, while MCB3 has $w = (\delta_0, 0) \neq 0$. This is a concrete obstruction for the actual supported-zero action and this specified norm. It does not prove failure of every topology or every arithmetic heat construction.

MCB3. Complete the graph without removing its mass. Retain the exact graph norm and embedding

$$\|(a, b)\|_E^2 = \|a\|_2^2 + 2\|b\|_2^2 + |m(a)|^2, \quad j(a, b) = (a, b, m(a)). \quad (\text{MCB4})$$

Its completion is precisely

$$\widehat{\mathcal{V}}_E = \ell^2(\mathbb{Z}) \oplus \mathbb{C}^{L \setminus \{1_L\}} \oplus \mathbb{C}. \quad (\text{MCB5})$$

The norm on MCB5 is $\|(a, b, m)\|_E^2 = \|a\|_2^2 + 2\|b\|_2^2 + |m|^2$. The factor two retains the two occurrences of b in the literal graph $((a, b), (m(a)\delta_0, b))$; none is discarded. Here the last coordinate is an independently retained mass; it is not defined by summing an arbitrary ℓ^2 sequence. To prove density, fix (a, b, m) in MCB5 and first truncate a on finite sets to $a^{[j]} \rightarrow a$ in ℓ^2 . Put $d_j = m - m(a^{[j]})$. Choose a finite set of integers disjoint from the truncation, of size $N_j > j^2 |d_j|^2$,

and put d_j/N_j on each of those coordinates, zero elsewhere. Call this finite correction $r^{[j]}$. It has sum d_j and squared norm $|d_j|^2/N_j < j^{-2}$. Hence

$$j(a^{[j]} + r^{[j]}, b) \longrightarrow (a, b, m). \quad (\text{MCB6})$$

The embedding is isometric by MCB4, and MCB5 is complete, proving the assertion. Equivalently the closure of the original operator graph in the product of the original coefficient Hilbert spaces is precisely

$$\{((a, b), (m\delta_0, b)) : a \in \ell^2(\mathbb{Z}), b \in \mathbb{C}^{L \setminus \{1_L\}}, m \in \mathbb{C}\}. \quad (\text{MCB20})$$

It is a closed linear relation: the displayed set is closed, and MCB6 proves density of the original graph in it. Its values at zero are exactly $(m\delta_0, 0)$. MCB5 retains this relation as a new state space; it does not falsely call it a single-valued closure on the old coefficient space.

Supported-zero multiplication now has the unique bounded extension

$$\widehat{E}(a, b, m) = (m\delta_0, b, m), \quad \widehat{E}^2 = \widehat{E}. \quad (\text{MCB7})$$

The formula agrees with MCB2 on the dense graph. Its output norm squared is $2|m|^2 + 2\|b\|_2^2$, at most twice the input norm squared, proving boundedness. The input $(0, 0, 1)$ attains the ratio two of squared norms, so $\|\widehat{E}\| = \sqrt{2}$. Idempotence follows by substitution. The projection forgetting mass has the exact sequence

$$0 \longrightarrow \mathbb{C} \xrightarrow{m \mapsto (0, 0, m)} \widehat{V}_E \xrightarrow{(a, b, m) \mapsto (a, b)} \ell^2(\mathbb{Z}) \oplus \mathbb{C}^{L \setminus \{1_L\}} \longrightarrow 0. \quad (\text{MCB8})$$

Every map and its kernel is visible in the formula. Moreover the kernel is not invariant under $\widehat{E}$: $\widehat{E}(0, 0, 1) = (\delta_0, 0, 1)$. Thus the missing mass coordinate is required to receive this arithmetic action; the action cannot descend through that projection.

MCB4. Every zero-lattice multiplication is retained. For $\sigma \in L$, multiplication by $z_\sigma = (0, \sigma)$ on the original finite point source sends every top integer to z_σ , and every lower z_λ to $z_{\sigma \wedge \lambda}$. For $\sigma < 1_L$ its bounded extension on MCB5 is

$$\begin{aligned} \widehat{E}_\sigma(a, b, m) &= (0, b', 0), \\ b'_\nu &= m\mathbf{1}_{\nu=\sigma} + \sum_{\substack{\lambda < 1_L \\ \sigma \wedge \lambda = \nu}} b_\lambda, \quad \nu < 1_L, \end{aligned} \quad (\text{MCB9})$$

and $\widehat{E}_{1_L} = \widehat{E}$. Each sum is finite. Cauchy–Schwarz bounds each coordinate by a constant depending only on $|L|$ times $|m| + \|b\|_2$, so these are bounded. They agree with original point multiplication on the dense graph, which also proves the exact composition law

$$\widehat{E}_\sigma \widehat{E}_\lambda = \widehat{E}_{\sigma \wedge \lambda}. \quad (\text{MCB10})$$

The same equality follows by grouping the finite fibres of the meet maps in MCB9. In particular τ acts here by placing the total $m + \sum b_\lambda$ at its own lower coordinate; it is not identified with the supported-zero output of MCB7.

For every nonzero original integer k , the multiplication operator is also bounded:

$$\widehat{M}_k(a, b, m) = (P_k a, b, m), \quad (P_k a)_j = \begin{cases} a_{j/k} & k \mid j, \\ 0 & k \nmid j. \end{cases} \quad (\text{MCB11})$$

It is an isometry, since $n \mapsto kn$ is injective. On finite sequences it preserves the exact mass. Composition reproduces multiplication of nonzero integers, and its compositions with MCB9 reproduce every original product with a zero label, first on the dense graph and then by

continuity. The original support semilattice has therefore survived the completion as explicit bounded operators.

Their norms and faithfulness can also be evaluated exactly. For $\sigma < 1_L$ write $k_{\sigma,\nu} = \#\{\lambda < 1_L : \sigma \wedge \lambda = \nu\}$. In the original weighted norm MCB5, each coordinate functional b'_ν in MCB9 has squared dual norm $k_{\sigma,\nu}/2 + \mathbf{1}_{\nu=\sigma}$. Distinct rows use disjoint input coordinates, and the output has weight two at every lower coordinate. Cauchy–Schwarz in each disjoint row, with equality in a maximal row, therefore proves

$$\|\widehat{E}_\sigma\|^2 = \max_{\nu < 1_L} \{k_{\sigma,\nu} + 2\mathbf{1}_{\nu=\sigma}\}, \quad \|\widehat{E}_{0_L}\|^2 = |L| + 1. \quad (\text{MCB21})$$

No source norm has been replaced. Finally applying each operator to the embedded integer one, $(\delta_1, 0, 1)$, gives the distinct embedded point z_σ . Thus the representation of the full finite meet semilattice is faithful.

MCB5. The heat embedding has quantitative inverse bounds. Use the exact Gaussian k_T and Fourier convention of LH4–5, with $T > 0$ unchanged. On finite sequences let

$$A_T a(x) = \sum_{n \in \mathbb{Z}} a_n k_T(x - n), \quad P_a(\theta) = \sum_n a_n e^{-2\pi i n \theta}. \quad (\text{MCB12})$$

Direct integration of products of Gaussians gives

$$\|A_T a\|_{L^2(\mathbb{R})}^2 = \frac{1}{\sqrt{8\pi T}} \sum_{n,m} a_n \bar{a}_m e^{-(n-m)^2/(8T)} = \int_{-1/2}^{1/2} W_T(\theta) |P_a(\theta)|^2 d\theta, \quad (\text{MCB13})$$

where

$$W_T(\theta) = \sum_{j \in \mathbb{Z}} e^{-8\pi^2 T(\theta+j)^2}. \quad (\text{MCB14})$$

For the second equality integrate the finite trigonometric polynomial against this positive, uniformly convergent sum and unfold to the real Gaussian Fourier integral. Its $(n-m)$ -th Fourier coefficient is exactly the first coefficient in MCB13, by the same Gaussian transform LH5. Thus no unrecorded measure factor is involved. Since $|\theta| \leq 1/2$,

$$e^{-2\pi^2 T} \leq W_T(\theta) \leq 1 + 2 \sum_{j \geq 1} e^{-8\pi^2 T(j-1/2)^2} =: U_T < \infty. \quad (\text{MCB15})$$

Orthogonality of the finite exponentials in MCB13 now proves

$$e^{-2\pi^2 T} \|a\|_2^2 \leq \|A_T a\|_2^2 \leq U_T \|a\|_2^2. \quad (\text{MCB16})$$

Completion extends A_T uniquely to $\ell^2(\mathbb{Z})$, with closed range and bounded inverse on that range. To prove closedness directly, a convergent output sequence is Cauchy, the lower bound makes its inputs Cauchy, and the upper bound identifies the output of their limit. The inverse bound is the square root of the reciprocal lower bound, with the original T retained.

MCB6. A faithful heat state with active supported zero. Let the labelled heat output keep the marked mass coordinate:

$$\mathbb{H}_T(a, b, m) = \left(A_T a v_{1_L} + k_T \sum_{\lambda < 1_L} b_\lambda v_\lambda, m \right). \quad (\text{MCB17})$$

Its function component lies in $L^2(\mathbb{R}; \mathbb{C}^L)$. The lower-coordinate norm is exactly $\|k_T\|_2^2 \|b\|_2^2 = (8\pi T)^{-1/2} \|b\|_2^2$. Together with MCB16 and the unchanged last coordinate, this proves that $\mathbb{H}_T$ is

a bounded isomorphism onto its closed image. In the literal graph norm the squared lower and upper constants are respectively

$$\min\{e^{-2\pi^2 T}, (8\pi T)^{-1/2}/2, 1\}, \quad \max\{U_T, (8\pi T)^{-1/2}/2, 1\}.$$

Supported-zero action there is exactly

$$E_T \mathbb{H}_T(a, b, m) = \left(mk_T v_{1_L} + k_T \sum_{\lambda < 1_L} b_\lambda v_\lambda, m \right), \quad E_T \mathbb{H}_T = \mathbb{H}_T \hat{E}. \quad (\text{MCB18})$$

This is a bounded idempotent, by MCB7 and MCB17. All lower $\hat{E}_\sigma$ and nonzero integer actions similarly have the exact conjugates through the specified isomorphism; their formulas are MCB9 and MCB11 followed by MCB17.

For $U > 0$, convolution of the function component by k_U and retention of m gives the exact heat evolution

$$(k_U *, \text{id}) \mathbb{H}_T = \mathbb{H}_{T+U}. \quad (\text{MCB19})$$

It follows first on finite sources from $k_U * k_T = k_{U+T}$, and on the completed space by the bounds just proved. The actual endpoint Gaussian mk_T changes with time, whereas its retained mass stays m . Equation MCB18 shows why the mass cannot be forgotten in this evolution if the e -action is to remain defined.

MCB7. Relation to the theta comb and scope of the repair. The infinite integer comb of LH8 is a tempered source, but its top coefficients are not ℓ^2 , and its total mass is not a finite number. It is therefore outside MCB5. LH8–15 retain that source in the explicitly different tempered domain, where its marked zero and its original zeta Mellin receiver remain defined. Multiplication by e on finite truncations of that comb produces $(2N+1)\delta_0 v_{1_L}$, which has no finite distributional limit: test against a function of value one at zero. Thus MCB7 does not silently assign a finite mass to the whole comb.

There is nevertheless an exact common domain: finite atomic sources map by LH2 into tempered distributions and by MCB4 into the graph completion. With pr_{fun} denoting the function component, their exact common-domain identity is

$$\text{pr}_{\text{fun}} \mathbb{H}_T j(v) = \mathcal{H}_T \iota(v), \quad v \in \mathcal{V}_0.$$

It holds coefficientwise by LH4 and MCB17. The additional datum in the completed heat state is the explicitly calculated boundary mass, and MCB8 records the projection which loses it. This construction proves a precise inadequacy of the coefficient Hilbert quotient for the supported-zero action and supplies an explicit repair. It neither discards the original datum nor claims that failure of this particular quotient decides RH.

6 The retained mass enters the actual prime-boundary receiver

MRT1. The two source moments and their prime class. The preceding construction has an arithmetic receiver, which we now calculate rather than identifying the two source spaces by analogy. Use the even Schwartz space $H = \mathcal{S}(\mathbb{R})^{\text{even}}$, with the two original moments

$$m_H(h) = \left(h(0), \int_{\mathbb{R}} h(y) dy \right), \quad H_0 = \ker m_H, \quad R = \mathbb{C}[\mathbb{Q}_{>0}^\times], \quad I = \ker(\text{aug} : R \rightarrow \mathbb{C}). \quad (\text{MRT1})$$

The group basis elements obey $t_a t_b = t_{ab}$, without changing the rational dilation parameter. The actual moment-zero source and its algebraic coinvariants are

$$M_0 = \ker(\text{aug} \otimes m_H), \quad Q_0 = M_0 / IM_0, \quad \mathcal{B} = \bigoplus_{p \text{ prime}} \mathbb{C} \ell_p \otimes \mathbb{C}^2. \quad (\text{MRT2})$$

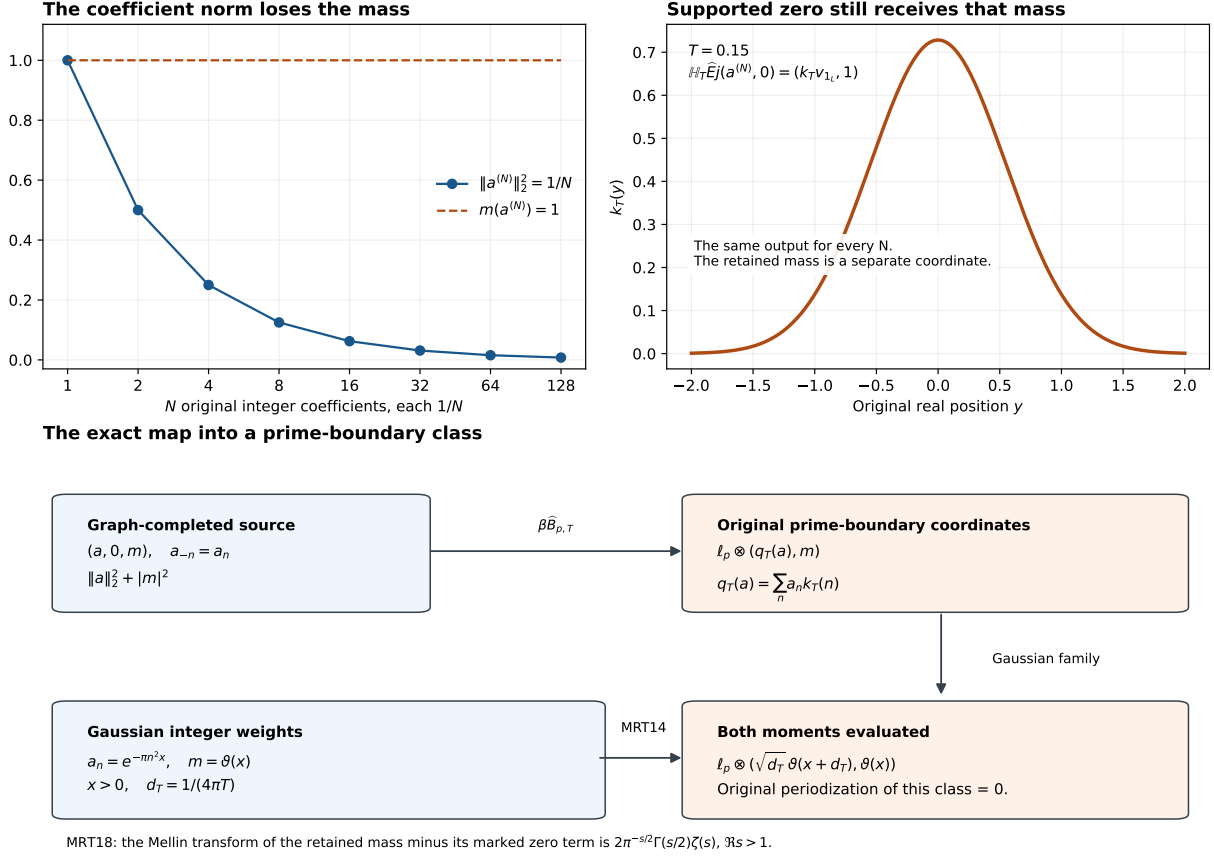


Figure 2: Retaining the mass repairs the specified supported-zero action. The left graph gives the exact coefficient norm of MCB3; the right graph shows its identical supported-zero heat output, at the stated time. The diagram records the actual prime-boundary map MRT7–12, whose original periodization seminorm remains zero. The Gaussian source MRT13–20 evaluates both boundary coordinates and returns the original zeta function. Curves are numerical renderings of the displayed exact formulas.

These are the original periodization source and prime-boundary coordinates of the published [original-zeta return, FR15–25](#). The periodization and its measure come from [Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*](#), Section III (6)–(19). The following explicit coefficient map and its mass-completion extension are derived here; they are not attributed to that article.

Here is the full algebra needed for this receiver. The functions

$$h_0(y) = (1 - 2\pi y^2)e^{-\pi y^2}, \quad h_1(y) = 2\pi y^2 e^{-\pi y^2}$$

have moments $(1, 0)$ and $(0, 1)$, respectively. Gaussian integration and evaluation prove these values. They give $H = H_0 \oplus \text{span}(h_0, h_1)$, so

$$Q_0 \cong H_0 \oplus (I/I^2) \otimes \mathbb{C}^2 \cong H_0 \oplus \mathcal{B}. \quad (\text{MRT3})$$

Indeed $M_0 = (R \otimes H_0) \oplus (I \otimes \mathbb{C}^2)$; its multiplication by I has components $I \otimes H_0$ and $I^2 \otimes \mathbb{C}^2$. In I/I^2 , the identity $t_{ab} - 1 = (t_a - 1) + (t_b - 1) + (t_a - 1)(t_b - 1)$, followed by the prime factorization of a positive rational, expresses every class as $\sum_p v_p(a)[t_p - 1]$. The linear functionals $t_a \mapsto v_p(a)$ kill I^2 and separate these generators. This proves both isomorphisms and their inverses.

Explicitly the coordinates of $c = [\sum_a t_a \otimes h_a]$ are

$$\Phi c = \sum_a h_a, \quad \beta(c) = \sum_p \ell_p \otimes \sum_a v_p(a) m_H(h_a). \quad (\text{MRT4})$$

The defining source condition gives $\Phi c \in H_0$. An inverse to MRT3 sends $(h, \sum_p \ell_p \otimes u_p)$ to

$$[1 \otimes h] + \sum_p [(t_p - 1) \otimes (u_{p,0} h_0 + u_{p,1} h_1)]. \quad (\text{MRT5})$$

There are only finitely many nonzero u_p . Changing the moment lift changes this class by IM_0 . These calculations verify all quotient relations; the receiver is not merely a map on arbitrarily selected representatives.

MRT2. The heat coefficient source has an exact two-moment map. Take an even finitely supported coefficient sequence a , with $a_{-n} = a_n$, no lower coefficients for this specified arrow, and the unchanged $T > 0$. Its heat function $h = A_T a$ of MCB12 belongs to H . Direct evaluation and integration give

$$q_T(a) = \sum_n a_n k_T(n), \quad m(a) = \sum_n a_n, \quad m_H(A_T a) = (q_T(a), m(a)). \quad (\text{MRT6})$$

Consequently, for every original arithmetic prime p , the map

$$B_{p,T}(a) = [(t_p - 1) \otimes A_T a] \in Q_0$$

has the exact coordinates

$$(\Phi, \beta) B_{p,T}(a) = (0, \ell_p \otimes (q_T(a), m(a))). \quad (\text{MRT7})$$

The first zero follows by subtraction of the same function, and the second follows from $v_p(p) = 1$, $v_p(1) = 0$. This is the connecting map from the actual integer heat source to the existing prime-boundary source.

Write $k_0 = k_T(0)$, $k_1 = k_T(1)$; these symbols retain the values $(4\pi T)^{-1/2}$ and $(4\pi T)^{-1/2} e^{-1/(4T)}$. For any pair (u_0, u_1) , put

$$a_0 = \frac{u_0 - k_1 u_1}{k_0 - k_1}, \quad a_1 = a_{-1} = \frac{k_0 u_1 - u_0}{2(k_0 - k_1)}, \quad a_n = 0 \quad (|n| > 1). \quad (\text{MRT8})$$

Since $k_0 > k_1$, this is defined. Substitution gives $q_T(a) = u_0$ and $m(a) = u_1$. Thus this actual finite source attains both prime-boundary moments, with their original coefficients and Gaussian constants. It is not confined to a single prescribed moment ratio.

MRT3. Completion and supported-zero action commute with this map. Let $\widehat{V}_E^{\text{even, top}}$ be the closed subspace $(a, 0, m)$ of MCB5 with even a . The samples $(k_T(n))_n$ lie in ℓ^2 ; hence Cauchy–Schwarz gives

$$|q_T(a)| \leq \left(\sum_n k_T(n)^2 \right)^{1/2} \|a\|_2. \quad (\text{MRT9})$$

The same formula with an independent last coordinate extends MRT7:

$$\widehat{B}_{p,T}(a, 0, m) = b(\ell_p \otimes (q_T(a), m)), \quad (\text{MRT10})$$

where b is the explicit map MRT5 on its second summand. The two scalar coordinates are continuous in the literal graph norm. No new norm is assigned to the adelic quotient in this statement. Even finite graph sources are dense in this subspace: in MCB6 choose the correcting sets in opposite pairs, retaining the same total correction and its vanishing squared norm.

The original supported-zero operator now has the following arithmetic receiving matrix, with all constants retained:

$$J_T = \begin{pmatrix} 0 & k_T(0) \\ 0 & 1 \end{pmatrix}, \quad J_T^2 = J_T, \quad \widehat{B}_{p,T} \widehat{E} = b(\text{id}_{\mathbb{C}\ell_p} \otimes J_T) \beta \widehat{B}_{p,T}. \quad (\text{MRT11})$$

Indeed $\widehat{E}(a, 0, m) = (m\delta_0, 0, m)$, whose two moments are $(k_T(0)m, m)$. This proves MRT11 exactly; it does not assert a not-yet-defined multiplication on all of Q_0 . On the completed pure-mass vector $(0, 0, 1)$, MRT10 gives $b(\ell_p \otimes (0, 1))$; after $\widehat{E}$ it gives $b(\ell_p \otimes (k_T(0), 1))$. The boundary action is consequently visible in the actual two-moment coordinates.

MRT4. Relation to the original periodization seminorm. Keep exactly the original periodization and its measure:

$$\mathcal{E}(c)(x) = x^{1/2} 2 \sum_{n \geq 1} (\Phi c)(nx), \quad \|c\|_\delta^2 = \int_0^\infty |\mathcal{E}(c)(x)|^2 (1 + \log^2 x)^{\delta/2} \frac{dx}{x}, \quad \delta > 1. \quad (\text{MRT12})$$

Connes’s compact-unit measure has mass one in this formula; with mass c_K the whole displayed integral has the factor c_K , which is retained. For every vector in MRT10, $\Phi = 0$, so $\mathcal{E} = 0$ and this exact seminorm is zero. Whenever $(q_T(a), m) \neq (0, 0)$, MRT3 proves that its boundary class is nonzero. This verifies on the actual Gaussian coefficient source the failure of its two moments to descend through the Hausdorff quotient of MRT12. The graph completion MCB5 repairs the domain of supported-zero multiplication; it does not turn MRT12 into a positive norm on these prime classes. The maps MRT7–12 retain both facts and give their exact connection.

MRT5. The theta family is an admitted source, with two evaluated moments. Let $x > 0$, independently of the heat time T , and put

$$a_n^x = e^{-\pi n^2 x}, \quad \vartheta(x) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x}, \quad d_T = \frac{1}{4\pi T}, \quad k_T(0) = \sqrt{d_T}. \quad (\text{MRT13})$$

These original weights are even and absolutely summable. Their finite truncations converge in the graph norm to $(a^x, 0, \vartheta(x))$: both the squared tails and the mass tails tend to zero by Gaussian convergence. Unlike the unweighted integer comb, this is an admitted vector of MCB5.

Moreover $A_T a^x$ is an even Schwartz function. To see every Schwartz bound directly, each derivative of $k_T(y - n)$ is a fixed polynomial in $y - n$ times its Gaussian. For any integer $M \geq 0$, use $1 + |y| \leq (1 + |n|)(1 + |y - n|)$. The supremum of the resulting polynomial times that Gaussian is finite, and its sum against $(1 + |n|)^M e^{-\pi n^2 x}$ is finite. Uniform derivative convergence proves

the assertion. Hence MRT7 applies directly to this actual Schwartz function, not just to its completed image. Evaluation and integration in MRT6 yield

$$\beta B_{p,T}(a^x) = \ell_p \otimes \left(\sqrt{d_T} \vartheta(x + d_T), \vartheta(x) \right). \quad (\text{MRT14})$$

The product of the two Gaussian factors proves the first entry, and integration of each translate proves the second. After the supported-zero action the same class has moments

$$\ell_p \otimes \left(\sqrt{d_T} \vartheta(x), \vartheta(x) \right). \quad (\text{MRT15})$$

Thus the change in this prime-boundary coordinate, in the order before minus after, is exactly

$$\ell_p \otimes \left(\sqrt{d_T} [\vartheta(x + d_T) - \vartheta(x)], 0 \right). \quad (\text{MRT16})$$

It is nonzero for every $x, T > 0$, since every pair of nonzero integer Gaussian terms strictly decreases when its parameter increases by $d_T > 0$. This evaluates an actual arithmetic consequence of the supported-zero operation in the retained source.

MRT6. Return to the original zeta function, with the marked term kept. The single integer-zero source is δ_0 , with coefficient one, mass one and heat moments $(\sqrt{d_T}, 1)$. Keep it as a separate summand. The remaining pair from MRT14 is

$$\left(\sqrt{d_T} [\vartheta(x + d_T) - 1], \vartheta(x) - 1 \right). \quad (\text{MRT17})$$

For $\Re s > 1$, its second Mellin integral is exactly

$$\int_0^\infty [\vartheta(x) - 1] x^{s/2-1} dx = 2\pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \zeta(s) = \frac{\pi^{s/2}}{2\Gamma(s/2)} \int_0^\infty [\vartheta(x) - 1] x^{s/2-1} dx. \quad (\text{MRT18})$$

Indeed the paired integers $\pm n$ give the factor two; the substitution $y = \pi n^2 x$ gives $\pi^{-s/2} n^{-s} \Gamma(s/2)$; absolute interchange follows from $\sum n^{-\Re s} < \infty$. The marked constant has no Mellin convergence half-plane on $(0, \infty)$ and remains separate, rather than being assigned zero. Poisson reflection and splitting at one retain two distinct endpoint contributions: the zero source coefficient gives $-1/s$, and the zero Fourier coefficient gives $1/(s-1)$, as derived in LH15. Thus original ζ and all its continuation factors remain explicit.

The first coordinate also has an evaluated Mellin transform: for $\Re s > 0$,

$$\int_0^\infty \sqrt{d_T} [\vartheta(x + d_T) - 1] x^{s/2-1} dx = 2\sqrt{d_T} \pi^{-s/2} \Gamma(s/2) \sum_{n \geq 1} e^{-\pi d_T n^2} n^{-s}. \quad (\text{MRT19})$$

Here the same calculation is absolutely convergent already for $\Re s > 0$, because the Gaussian coefficient controls the integer sum. The full Dirichlet series in MRT19 is entire in s , by uniform Gaussian convergence on compact sets; the Gamma factor is retained. In the common half-plane $\Re s > 1$, the Mellin transform of MRT16 is therefore

$$2\sqrt{d_T} \pi^{-s/2} \Gamma(s/2) \left[\sum_{n \geq 1} e^{-\pi d_T n^2} n^{-s} - \zeta(s) \right] \ell_p \otimes (1, 0). \quad (\text{MRT20})$$

Every term in this difference has now been identified. In particular the zero-seminorm prime-boundary source contains a family whose retained mass returns the original zeta function. This is a proved receiver and a proved loss of specified data under MRT12; it is not an evaluation of the full Weil pairing on a newly invented norm.

7 The endpoint obstruction defines a two-dimensional heat boundary

ESH1. The original theta density before its differential filter. Keep the original coordinate $s = \frac{1}{2} + \frac{iZ}{2}$, and the full factor $A(s) = \pi^{-s/2}\Gamma(s/2)$. The theta formula LH15, with the exact substitution $x = e^{4u}$, gives

$$A(s)\zeta(s) = \int_0^\infty \rho(u) \cos(Zu) du, \quad \rho(u) = 8e^u\psi(e^{4u}) - 4e^{-u}, \quad |\Im Z| < 1. \quad (\text{ESH1})$$

Indeed the two integral terms in LH15 become $8 \int_0^\infty e^u\psi(e^{4u}) \cos(Zu) du$. Its rational endpoint contribution is $1/[s(s-1)] = -4/(1+Z^2)$, equal to the retained $-4e^{-u}$ integral on the stated strip. Both endpoint terms are consequently present in ESH1. The density and the subsequent theta filter use the original definitions of [Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*](#), equations `phidef`, `htdef`, `hoz` and `sas`; the unfiltered sectorial extension is derived here.

The original unfiltered real-axis integral after inserting a heat weight is

$$I_t(Z) = \int_0^\infty e^{tu^2} \rho(u) \cos(Zu) du. \quad (\text{ESH2})$$

It converges for $\Re t < 0$, for every complex Z , locally uniformly together with all derivatives. The part $8e^u\psi(e^{4u})$ has faster than Gaussian decay, while the endpoint part has Gaussian decay precisely in this left half-plane. For positive real t and $Z = 0$, the latter part diverges with one sign. Thus the original real-axis formula does not define an entire-in- Z heat family at positive time. An analytic continuation must be specified rather than calling that same improper integral convergent.

ESH2. Exact lateral continuations of the endpoint. Set $q = -t$, first $q > 0$. For any complex a , completing the square gives

$$F(q, a) = \int_0^\infty e^{-qu^2 - au} du = \frac{\sqrt{\pi}}{2\sqrt{q}} e^{a^2/(4q)} \operatorname{erfc}\left(\frac{a}{2\sqrt{q}}\right). \quad (\text{ESH3})$$

For real a this is a direct real integral substitution; both sides are entire in a by Gaussian domination and the entire power series of erfc , so the identity holds for all a . The right side extends analytically to $q \in \mathbb{C} \setminus (-\infty, 0]$, with the square root positive on the positive real axis. This specifies the branch. Here $\operatorname{erfc}(z) = 1 - \frac{2}{\sqrt{\pi}} \int_0^z e^{-v^2} dv$; its entire primitive immediately proves $\operatorname{erfc}(z) + \operatorname{erfc}(-z) = 2$. This is the convention of N. M. Temme, [NIST DLMF, equation 7.2.2](#); the original equation TeX is retained with the source record.

The endpoint cosine integral is

$$B(q, Z) = \frac{1}{2} \{F(q, 1 - iZ) + F(q, 1 + iZ)\}. \quad (\text{ESH4})$$

For positive real t , define $B_+(t, Z)$ and $B_-(t, Z)$ by approaching $q = -t$ from the upper and lower q -half-planes, respectively. The square roots are $+i\sqrt{t}$ and $-i\sqrt{t}$. Thus ESH3 and the displayed erfc identity give the exact difference

$$F_+(t, a) - F_-(t, a) = -i\sqrt{\frac{\pi}{t}} e^{-a^2/(4t)}, \quad B_+(t, Z) - B_-(t, Z) = -i\sqrt{\frac{\pi}{t}} e^{(Z^2-1)/(4t)} \cos \frac{Z}{2t}. \quad (\text{ESH5})$$

This orientation uses the q -plane; reversing the order of the two lateral values reverses the sign.

The rapidly decaying part

$$L_t(Z) = 8 \int_0^\infty e^{tu^2+u}\psi(e^{4u}) \cos(Zu) du$$

is jointly entire in (t, Z) . The two unfiltered continuations at positive time are therefore

$$I_{\pm}(t, Z) = L_t(Z) - 4B_{\pm}(t, Z), \quad I_+ - I_- = 4i\sqrt{\frac{\pi}{t}} e^{(Z^2-1)/(4t)} \cos \frac{Z}{2t}. \quad (\text{ESH6})$$

The factor four and its sign come from the original endpoint density in ESH1. Each branch satisfies $\partial_t I = -\partial_Z^2 I$: differentiate the integral in $\Re t < 0$, then use analytic continuation on its specified branch. This continuation assertion does not make ESH2 converge at positive real time.

ESH3. The full theta filter and its exact kernel. Define, without suppressing its time-dependent terms,

$$\mathcal{N}_t = -\frac{1 + (Z - 2t\partial_Z)^2}{64} = -\frac{1 + Z^2 - 2t - 4tZ\partial_Z + 4t^2\partial_Z^2}{64}. \quad (\text{ESH7})$$

For a twice differentiable density with the decay of ρ , two integrations by parts in $\Re t < 0$ give

$$\int_0^\infty e^{tu^2} \rho''(u) \cos(Zu) du = -\rho'(0) - (Z - 2t\partial_Z)^2 I_t(Z). \quad (\text{ESH8})$$

The first boundary at infinity vanishes; at zero it is $-\rho'(0)$. The second boundary is zero because the derivative of $e^{tu^2} \cos(Zu)$ vanishes at zero. Differentiating that whole weight twice gives precisely $(2t + 4t^2u^2 - Z^2) \cos(Zu) - 4tZu \sin(Zu)$, which verifies every term of ESH8.

The Poisson identity in LH14 yields $\psi'(1) = -\psi(1)/4 - 1/8$. Consequently

$$\rho'(0) = 8\psi(1) + 32\psi'(1) + 4 = 0. \quad (\text{ESH9})$$

There has been no dropped endpoint boundary. Direct differentiation of the entire density, with $x = e^{4u}$, gives

$$\frac{\rho''(u) - \rho(u)}{64} = \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}} = \Phi(u). \quad (\text{ESH10})$$

The $-4e^{-u}$ term is annihilated by $\partial_u^2 - 1$, but its boundary contribution has already been included in ESH9. ESH8–10 prove, first in $\Re t < 0$ and then along each continuation, the exact identity

$$\mathcal{N}_t I_+(t, Z) = \mathcal{N}_t I_-(t, Z) = H_t(Z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(Zu) du. \quad (\text{ESH11})$$

The integral for H_t is jointly entire by the decay of Φ . This is a derived equality between the two specified heat constructions, including their lost boundary directions. The two density parts separately have nonzero boundary terms:

$$\mathcal{N}_t B = -\frac{1}{64}, \quad \mathcal{N}_t L_t = H_t - \frac{1}{16}, \quad \mathcal{N}_t (L_t - 4B) = H_t. \quad (\text{ESH17})$$

For the first equation the endpoint density satisfies $\rho'' = \rho$, $\rho'(0) = -1$, so ESH8 gives it directly. For the second, the theta-only density has derivative -4 at zero by ESH9 and the endpoint derivative $+4$ in the full density. Its interior filtered density is still Φ . Substitution in ESH8 yields the displayed $-1/16$, which the endpoint contribution cancels exactly. Thus annihilating the interior endpoint density does not authorize omitting its integration boundary.

For each $t \neq 0$, the kernel of $\mathcal{N}_t$ on the entire functions of Z is exactly

$$\left\{ e^{Z^2/(4t)} \left(c_+ e^{iZ/(2t)} + c_- e^{-iZ/(2t)} \right) : c_+, c_- \in \mathbb{C} \right\}. \quad (\text{ESH12})$$

To prove completeness of this list, write $f = e^{Z^2/(4t)} g$. Then $(Z - 2t\partial_Z)f = -2te^{Z^2/(4t)} g'$, and the kernel equation becomes $g + 4t^2 g'' = 0$. Its two initial values determine its entire solution by

the coefficient recursion, giving exactly the two displayed exponentials. ESH5 lies in this space, so its annihilation in ESH11 is also verified directly. The actual cosine source is even in Z . On this original even domain the kernel is one-dimensional: evenness in ESH12 requires $c_+ = c_-$, leaving $e^{Z^2/(4t)} \cos(Z/(2t))$. The sine direction belongs to the explicitly larger, nonsymmetric entire-function domain. Thus the actual lateral jump occupies exactly the symmetric kernel direction, not two independently varied cosine parameters. The time dependence of any such homogeneous difference which also satisfies the heat equation is determined. Writing the two coefficients in ESH12 as functions $c_{\pm}(t)$, direct differentiation gives

$$c'_{\pm}(t) + \left(\frac{1}{2t} - \frac{1}{4t^2} \right) c_{\pm}(t) = 0, \quad c_{\pm}(t) = C_{\pm} t^{-1/2} e^{-1/(4t)}. \quad (\text{ESH18})$$

On a fixed simply connected time domain avoiding zero the chosen square root makes this formula single-valued. The two independent space exponentials force both scalar equations, and integrating them gives exactly the stated constants. In original s coordinates the modes are $t^{-1/2} e^{-s^2/t}$ and $t^{-1/2} e^{-(s-1)^2/t}$. The even source requires equal coefficients, recovering the time dependence of ESH6 and ESH14.

ESH4. Both original-zeta receivers and the endpoint modes. Return to the original coordinate $Z = -2i(s - 1/2)$ and keep

$$f_{\pm}(t, s) = \frac{I_{\pm}(t, -2i(s - 1/2))}{\pi^{-s/2} \Gamma(s/2)}, \quad z_t(s) = \frac{16H_t(-2i(s - 1/2))}{s(s - 1)\pi^{-s/2} \Gamma(s/2)}. \quad (\text{ESH13})$$

The first family is defined on the sectorial branches just proved; the second is CF1's returned meromorphic family. Both recover the original $\zeta(s)$ at time zero on $0 < \Re s < 1$: for the first, approach zero through $t < 0$ and dominate the endpoint integral by $e^{-(1-|\Im Z|)u}$; for the second, use ESH1 and $16\mathcal{N}_0 = s(s - 1)$. Their domains at time zero and their later poles have not been identified merely from their common initial values.

The actual two endpoint modes in the original coordinate are $e^{-s^2/t}$ and $e^{-(s-1)^2/t}$. Specifically,

$$f_+(t, s) - f_-(t, s) = 2i \sqrt{\frac{\pi}{t}} \frac{\pi^{s/2}}{\Gamma(s/2)} \left[e^{-s^2/t} + e^{-(s-1)^2/t} \right]. \quad (\text{ESH14})$$

This follows by expanding the cosine in ESH6 and substituting the stated coordinate, with no rescaling of time.

The complete differential receiving map is

$$z_t(s) = \frac{[s(s - 1) + t/2]A(s)f_{\pm}(t, s) + \frac{t(2s-1)}{2}\partial_s[A(s)f_{\pm}(t, s)] + \frac{t^2}{4}\partial_s^2[A(s)f_{\pm}(t, s)]}{s(s - 1)A(s)}. \quad (\text{ESH15})$$

For verification, $Z - 2t\partial_Z = -i[(2s - 1) + t\partial_s]$; squaring retains the extra derivative term $2t$. Multiplication by 16 in ESH7 then gives exactly the numerator of ESH15. This proves the equality away from the displayed exceptional factors and hence meromorphically, with the original factor germs retained. The first derivative contributes $A'f + Af'$, and the second contributes $A''f + 2A'f' + Af''$, with

$$A'/A = -\frac{1}{2} \log \pi + \frac{1}{2} \psi_{\Gamma}(s/2), \quad A''/A = \frac{1}{4} \psi'_{\Gamma}(s/2) + [-\frac{1}{2} \log \pi + \frac{1}{2} \psi_{\Gamma}(s/2)]^2. \quad (\text{ESH16})$$

Here ψ_{Γ} denotes the Gamma logarithmic derivative, distinct from the theta sum $\psi(x)$. Every derivative of the original multiplier is therefore specified.

At fixed $t \neq 0$ on these branches, $I_{\pm}$ is entire in s , as is $1/A(s)$, so $f_{\pm}$ is entire. In contrast, CF27 proves that z_t has a nonzero pole at one for real t . ESH15 is the exact map between these different analytic domains; its denominator and derivatives cannot be deleted. The recovery

of ζ 's pole as $t \rightarrow 0$ for the sectorial family is not locally uniform at that pole, since a locally uniform limit of holomorphic functions would be holomorphic there. This identifies the precise endpoint domain change, not a failure of the proved convergence in the open strip.

Thus the obstruction ESH2 defines both a two-branch continuation and the explicit two-dimensional entire kernel ESH12, whose intersection with the actual even source is one-dimensional. The map to the theta heat family annihilates exactly these homogeneous directions on the respective stated domains at each nonzero time. Keeping a chosen continued source and its endpoint coordinates preserves them; the subsequent map to the original ζ is ESH13–16 with all factors retained. No conclusion about RH follows merely from forgetting or retaining this computed kernel.